# Department of General Education

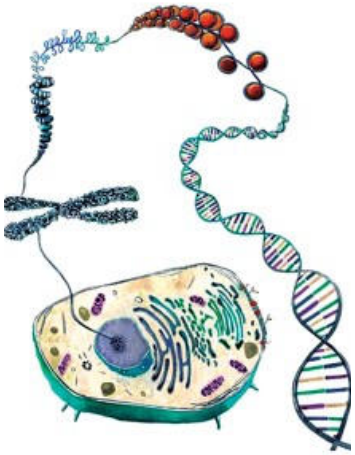
# Kollam District

## SMART A+ – 2026



## Biology

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# Unit 1

# Genetics of Life

Analyse the given ***Assertion*** and ***Reason*** and choose the correct answer.

1.**Assertion:** DNA is made up of double strands.
**Reason:** Nitrogen bases pair specifically through hydrogen bonds.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true

2.**Assertion:** The Nitrogen base Thymine is present in RNA.
**Reason:** The sugar present in RNA is ribose.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

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3.**Assertion:** Genes control protein synthesis.
**Reason:** Genes give instructions for the sequence of amino acids.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

4.**Assertion:** In humans, the father plays a role in sex determination.
**Reason:** The Y chromosome from father carries the SRY gene.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

5.**Assertion:** Recessive traits are not expressed in the first generation ($F_1$).
**Reason:** Dominant traits suppress recessive traits.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false

C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

6.**Assertion:** In the second generation ($F_2$) of a monohybrid cross, a 3:1 phenotypic ratio is obtained.
**Reason:** Alleles segregate during the formation of gametes.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

7.**Assertion:** All variations are beneficial for the survival of organisms.
**Reason:** Evolution cannot occur without variation.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

8.**Assertion:** Nucleotide is the basic unit of DNA.
**Reason:** Nucleotides exist independently without bonding with each other.

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A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

9.**Assertion:** In incomplete dominance, the first generation shows an intermediate character.
**Reason:** Neither allele shows complete dominance over the other.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

10.**Assertion:** rRNA plays a vital role in protein synthesis.
**Reason:** rRNA forms the structure of ribosomes and performs its function.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

11.**Assertion:** A gene can have more than two allelic forms in a population.
**Reason:** An individual can carry all the alleles present in a population.

A) Assertion is true, but Reason is false
B) Both Assertion and Reason are false
C) Both Assertion and Reason are true, but Reason is not the correct explanation
D) Both Assertion and Reason are true, and Reason is the correct explanation
E) Assertion is false, but Reason is true.

# Evaluate the statements and find the correct answer

## 12. Analyse the following statements related with DNA

i. DNA consists of two antiparallel strands
ii. DNA contains deoxy ribose sugar
iii. Adenine pairs with guanine
iv. Uracil is absent in DNA

Options:
A) i, ii true; iii, iv false
B) i, ii, iv true; iii false
C) ii, iii true; i, iv false
D) All are correct

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## 13. Evaluate the statements related with the functions of genes

i. Genes give instructions for protein synthesis
ii. Genes are segments of DNA
iii. All genes produce the same protein
iv. Changes in genes cause variations.

Options:
A) i, ii true; iii, iv false
B) i, ii, iv true; iii false
C) ii, iii true; i, iv false
D) All are correct

## 14. Read the statements on human chromosomes and answer the questions.

i. Humans have 23 pairs of chromosomes
ii. All chromosomes take part in sex determination
iii. X chromosome is larger than Y chromosome
iv. Autosomes control only sex determination.

Options:
A) i, iii true; ii, iv false
B) i, ii, iv true; iii false

C) ii, iii true; i, iv false
D) All are correct.

## 15. Analyse the statements pertaining RNA and choose the best option.

i. RNA is usually single-stranded
ii. Thymine is present in RNA
iii. RNA participates in protein synthesis
iv. RNA contains ribose sugar.

Options:
A) i, ii true; iii, iv false
B) i, ii, iv true; iii false
C) i, iii, iv true; ii false
D) All are correct.

## 16. Choose the correct option regarding Protein synthesis

i. Transcription occurs in the nucleus
ii. Translation occurs in ribosomes
iii. tRNA carries amino acids
iv. rRNA has no role in protein synthesis

Options:
A) i, ii true; iii, iv false
B) i, ii, iii true; iv false
C) ii, iii true; i, iv false
D) All are correct.

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## 17. Regarding Mendel's experiments,which answer is correct?

i. Each character is controlled by a pair of factors
ii. Recessive traits disappear permanently
iii. Dominant traits are expressed in $F_1$
iv. Alleles separate during gamete formation.

Options:
A) i, iii,iv true; ii false
B) i, iii, true; ii ,iv false
C) ii, iii true; i, iv false
D) All are correct.

### 18. Choose the right option regarding Heredity and variation

i. Heredity ensures continuity of traits
ii. Variations exist among individuals of the same species
iii. All variations are harmful
iv. Variation is essential for evolution

Options:
A) i, ii , iv true; iii false

B) i, ii, true; iii ,iv false
C) ii, iii true; i, iv false
D) All are correct.

**19. Find out the correct option regarding the structure DNA**

i. Nucleotides are the basic units of DNA
ii. A nucleotide consists of sugar, phosphate and nitrogen base
iii. Nitrogen bases pair randomly
iv. DNA is formed by joining nucleotides

Options:
A) i, ii true; iii, iv false
B) i, ii, iv true; iii false
C) ii, iii true; i, iv false
D) All are correct.

## 20. Analyse the statements regarding Phenotype and genotype and choose the correct answer.

i. Phenotype refers to visible traits
ii. Genotype refers to genetic constitution
iii. Genotype changes due to environment
iv. Environment influences phenotype

Options:
A) i, ii true; iii, iv false
B) i, ii, iv true; iii false
C) ii, iii true; i, iv false
D) All are correct.

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## 21. Evaluate the statements related to sex determination in humans and choose the correct answer.

i. Presence of Y chromosome determines male sex
ii. SRY gene causes testis development in embryo
iii. Mother determines the sex of the child
iv. Individuals with XX genotype are females

Options:
A) i, ii true; iii, iv false
B) i, iii true; i, iv false
C) i , ii, iv true; iii false
D) All are correct.

22. Analyse the statement and answer the questions. (2 marks)

**“The discovery of CRISPR-Cas9 technology is expected to bring revolutionary advances in gene therapy.”**

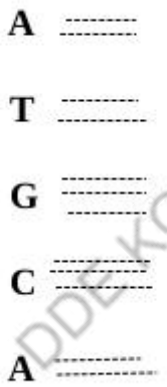
a) Name the scientists who shared the Nobel Prize for gene-editing research.
b) Mention two applications of CRISPR-Cas9 other than gene therapy.

23. Do you agree that DNA and RNA have the same basic structural units? Explain. (2 marks)

**24. Answer the questions regarding chromosome structure (2 marks)**

a) Name the two major chemical components of chromosomes.
b) What is the structure formed when DNA coils around a histone octamer?

**25.** Nitrogen bases of one DNA strand are given. Complete the diagram by writing the corresponding nitrogen bases of the RNA that can be formed from the given DNA Strand..
**(2 marks)**



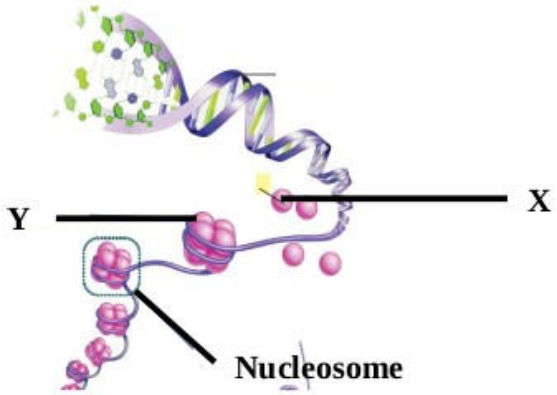

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26. a) How many pairs of chromosomes are present in a human somatic cell ( Body cell) ? (1 mark)
b) Why are somatic cell chromosomes called homologous chromosomes? (1 mark)

27. **A Biologist ,who follows Mendel's experiments, crossed a pure breeding tall pea plant(T T) with a dwarf one (t t). When he self pollinated the F1 offsprings , he got 400 plants ( offsprings) in the F2 generation.**

a) Approximately how many dwarf plants will appear in $F_2$ generation? (1 mark)
b) Why does the hidden trait reappear in the second generation? (1 mark)

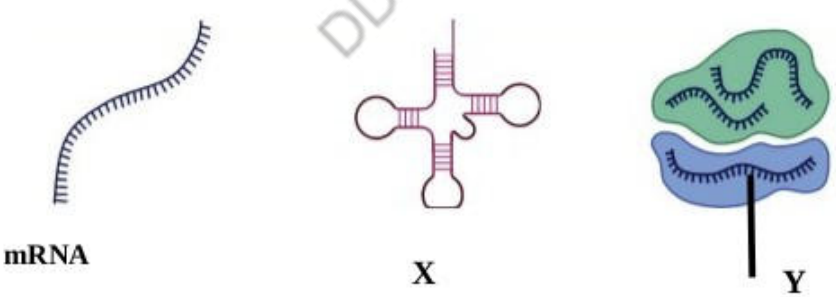
28. **Observe the given diagram and answer the questions.**




a) Identify parts X and Y. (1 mark)
b) How is a nucleosome formed? (1 mark)

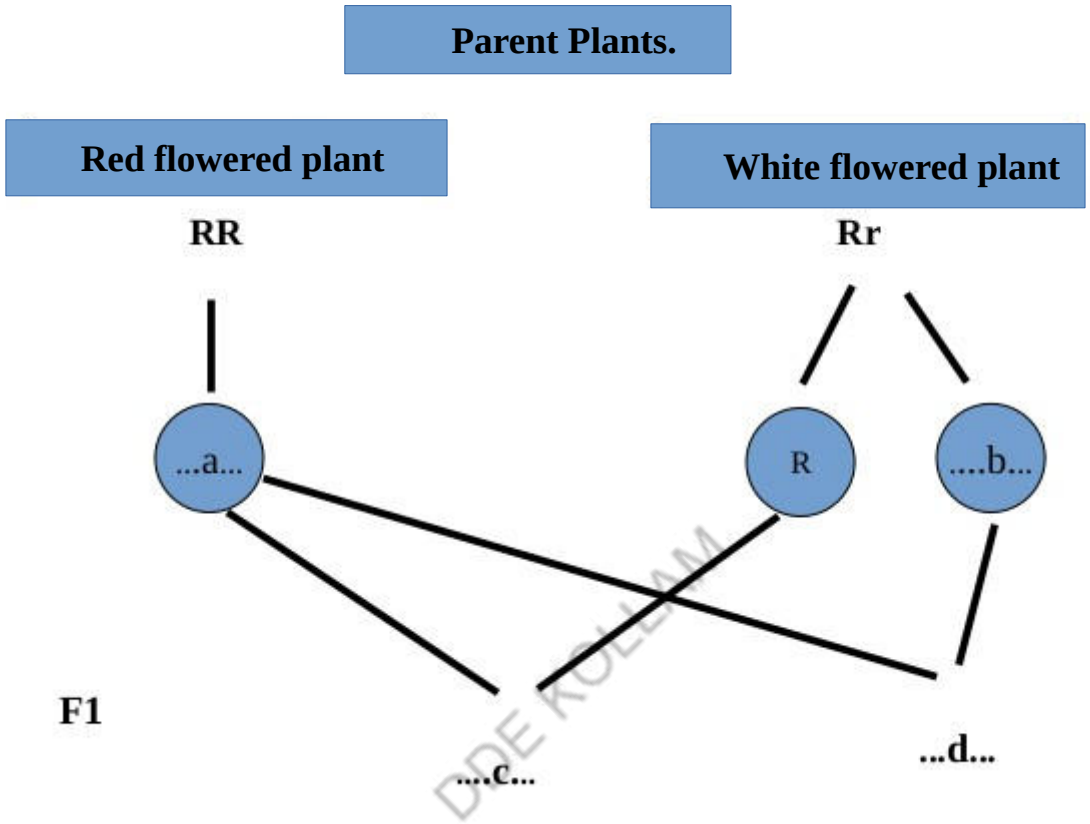
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29. Observe the diagrams and answer the questions.




a) Identify X and Y. (1 mark)
b) How will a structural defect to Y affect protein synthesis? (2 marks)

30. **Analyse the illustration and answer the questions.** **( 3 marks)**

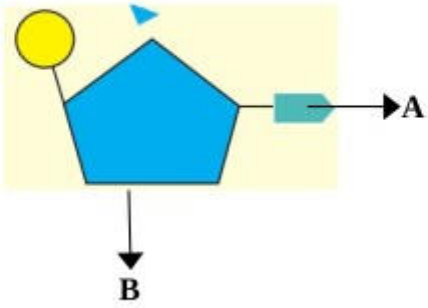



A) Identify the Alleles ' **a'** and **'b'.**

B) What would be the genotype of the offspring labelled as '**c'** in the **F1** generation.

C) What is the phenotype of the offspring labelled '**d'.**

**31. Observe the diagram of Nucleotide and answer the questions.** **(4 marks)**




a) Identify the labelled parts A & B
b) Which are the parts that differ between DNA and RNA?
c) Which component participates in the formation of bonds that link nucleotides together ?
d) Name the bond that is formed between nitrogen bases in DNA .

32. **Given below is a schematic diagram of protein synthesis**

DNA —Transcription (Nucleus)→ **X** —**Y** (Cytoplasm)→ Protein

1. a) Identify **X** and **Y**
2. b) What is the role of **X** in protein synthesis ?
3. c) Identify the role of tRNA and rRNA in protein synthesis

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**33. During protein synthesis, the information stored in DNA is not directly used at the ribosome' ( 4 marks)**

Analyse the statement and answer the following questions.

a) What are the processes that are involved in protein synthesis? (1 mark)

b) Name the molecule that carries information from DNA to ribosome ( 1 mark)

c) Explain the steps that take place in cytoplasm . ( 2 marks)

**34. A teacher explained in class that , at a particular stage during meiosis, two homologous chromosomes exchange segments with each other,resulting in new combinations of genes in the gametes.**

a) Name the process described above. (1 mark)

b) In which stage of meiosis does this process occur? (1 mark)

c) How does this process contribute to genetic variation? (2 marks)

# Biology

## Chapter 1.

## Answer Key

**Assertion–Reason (1–11)**

1. **D**
2. **E**
3. **D**
4. **D**
5. **D**
6. **D**
7. **E**
8. **A**
9. **D**
10. **D**
11. **A**

**Evaluate the Statements (12–21)**

12. **B**
13. **B**
14. **A**
15. **C**
16. **B**
17. **A**
18. **A**
19. **B**
20. **B**
21. **C**

**Very Short / Short Answer**

**22. CRISPR-Cas9**

a) Emmanuelle Charpentier, Jennifer Doudna

b) Cancer treatment, improving crop varieties.

**23.** Yes. Both have nucleotides as basic units (sugar, phosphate, nitrogen base).But there is difference in type of sugar( DNA: Deoxy ribose, RNA : Ribose) and Nitrogen base( DNA: Thymine, RNA: Uracil)

**24.**

a) DNA and proteins (histones)

b) Nucleosome

**25.** Complementary RNA bases:

DNA → RNA : A–U, T–A, G–C, C–G,A-U

**26.**

a) 23 pairs

b) They are similar in size, shape, and gene position and are received one from each parent .

**27.**

a) 100 dwarf plants ( 3:1 i e 300:100)

b) Due to segregation of alleles during gamete formation.

**Diagram-based**

**28.**

a) X – Histone, Y – Octamer.

b) DNA coils around histone octamer.

**29.**

a) X – tRNA, Y – rRNA

b) Protein synthesis will be affected/stop.( The joining of amino acids to form protein will be affected)

**30.**

a) a – R, b – r

b) c – RR

c) d – Red flower

**31. Nucleotide**

a) A – Nitrogen base, B – Sugar

b) Sugar and nitrogen base

c) Phosphate

d) Hydrogen bond

**Protein Synthesis**

**32.**

a) X – mRNA, Y – Translation

b) mRNA carries genetic code( carries message for protein Synthesis from DNA)

c) tRNA brings amino acids; rRNA forms ribosome which helps joining amino acids to form proteins.

**33.**

a) Transcription, Translation
b) mRNA
c) Translation occurs at ribosome using mRNA, tRNA

**Meiosis**

**34.**

a) Crossing over
b) Prophase I
c) Exchange of chromosome materials causes recombination of alleles causing genetic variation in offsprings.

# Chapter 2.

# Paths of Evolution.

## Part 1: Evaluate the Statements

**1. Evaluate the following statements regarding Lamarckism and choose the correct option.**

i. Characters acquired by organisms are inherited by the next generation.

ii. Use and disuse of organs affect evolution.

iii. Giraffes developed long necks through natural selection.

iv. Changes in the environment create new needs in organisms.

A) i, ii, iv are correct; iii is wrong B) i, iii are correct; ii, iv are wrong

C) ii, iii are correct; i, iv are wrong D) i, ii are correct; iii, iv are wrong

**2. Evaluate the statements regarding Antibiotic Resistance:**

i. Antibiotic resistance cannot be inherited.

ii. Antibiotic resistance is an acquired character.

iii. Resistant bacteria survive even in the presence of antibiotics.

iv. Mutations in bacteria enable them to resist antibiotics.

A) i, iii are correct; ii, iv are wrong B) i, ii, iii are correct; iv is wrong

C) iii, iv are correct; i, ii are wrong D) i, iv are correct; ii, iii are wrong

**3. Evaluate the statements regarding Superbugs:**

i. Superbugs are bacteria capable of resisting multiple antibiotics.

ii. Superbugs can be easily destroyed using normal antibiotics.

iii. Superbugs reduce the effectiveness of treatment.

iv. Superbugs are formed due to mutations.

A) i, ii, iii are correct; iv is wrong B) i, ii are correct; iii, iv are wrong

C) ii, iv are correct; i, iii are wrong D) i, iii, iv are correct; ii is wrong

**4. Evaluate the statements regarding Darwinism:**

i. The theory of Natural Selection was proposed by Darwin.

ii. Variations help organisms to survive.

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iii. Acquired characters are inherited.

iv. Overproduction leads to the struggle for existence.

A) i, ii, iv are correct; iii is wrong B) i, iii are correct; ii, iv are wrong

C) ii, iii, iv are correct; i is wrong D) i, ii, iii are correct; iv is wrong

**5. Evaluate the statements regarding the evolution of Giraffes:**

i. Long necks evolved due to the use and disuse of organs.

ii. Giraffes with long necks survived better.

iii. Giraffes with short necks reproduced more.

iv. Environmental pressure influenced evolution.

A) ii, iii are correct; i, iv are wrong B) i, ii, iii are correct; iv is wrong

C) i, ii, iv are correct; iii is wrong D) iii, iv are correct; i, ii are wrong

**6. Evaluate the statements regarding Galapagos Finches:**

i. All finches have the same type of beaks.

ii. Variation in beaks is related to food availability.

iii. Natural selection influenced the shape of the beak.

iv. Finches evolved into different species.

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A) i, iii, iv are correct; ii is wrong B) i, ii are correct; iii, iv are wrong

C) ii, iii, iv are correct; i is wrong D) iii, iv are correct; i, ii are wrong

**7. Evaluate the statements regarding Natural Selection:**

i. Organisms produce more offspring than can survive.

ii. Variations are inherited.

Iii. All offspring survive until maturity.

iv. Survival of the fittest occurs.

A) i, iii, iv are correct; ii is wrong B) ii, iii are correct; i, iv are wrong

C) i, ii are correct; iii, iv are wrong D) i, ii, iv are correct; iii is wrong

**8. Evaluate the statements regarding Speciation:**

i. Speciation leads to the formation of new species.

ii. Speciation is a process that happens very quickly.

iii. Geographical isolation can cause speciation.

iv. Accumulation of variations leads to speciation.

A) i, ii, iv are correct; iii is wrong B) i, iii, iv are correct; ii is wrong

C) ii, iii are correct; i, iv are wrong D) iii, iv are correct; i, ii are wrong

**9. Evaluate the statements regarding Fossils:**

i. Fossils are the remains of ancient organisms.

ii. Fossils provide evidence for evolution.

iii. Fossils act as links in the evolutionary process.

iv. Fossils are found only in living organisms.

A) i, ii, iii are correct; iv is wrong B) ii, iii are correct; i, iv are wrong

C) i, iii are correct; ii, iv are wrong D) iii, iv are correct; i, ii are wrong

**10. Evaluate the statements regarding Molecular Evidence:**

i. Molecular evidence is against the theory of evolution.

ii. There is a high similarity in DNA between humans and chimpanzees.

iii. DNA similarity indicates an evolutionary relationship.

iv. Comparison of amino acid sequences helps in the study of evolution.

A) i, ii, iv are correct; iii is wrong B) i, iii are correct; ii, iv are wrong

C) ii, iii, iv are correct; i is wrong D) iii, iv are correct; i, ii are wrong

**11. Evaluate the statements related to Lamarckism:**

i. Characters acquired by organisms during their lifetime are passed on to the next generation.

ii. Continuous use of an organ leads to its development and increased capability.

iii. Scientists have proven that acquired variations change the genetic structure of organisms.

iv. Environmental changes create new needs in organisms.

A) i, ii, iv are correct; iii is wrong B) i, iii are correct; ii, iv are wrong

C) i, ii are correct; iii, iv are wrong D) All statements are correct.

**12. Evaluate the statements regarding Darwinism (Theory of Natural Selection):**

i. Organisms produce more offspring than the environment can support (Overproduction).

ii. Competition between organisms for limited resources leads to the struggle for existence.

iii. All variations in organisms are always beneficial to them.

iv. Accumulation of favourable variations over generations leads to speciation.

A) i, ii are correct; iii, iv are wrong. B) i, ii, iv are correct; iii is wrong.

C) iii is correct; i, ii, iv are wrong. D) ii, iii are correct; i, iv are wrong.

**13. Evaluate the statements regarding Molecular Biology evidence:**

i. Comparing the sequence of nucleotides in DNA helps to find the evolutionary relationship between organisms.

ii. There is no difference in the amino acid sequence of the beta chain of haemoglobin in humans and chimpanzees.

iii. Similarity in the sequence of amino acids in proteins indicates a common ancestor.

iv. The organism with the closest evolutionary relationship to humans is the mouse.

A) i, ii, iv are correct; iii is wrong. B) iv only is correct.

C) i, ii are correct; iii, iv are wrong. D) i, ii, iii are correct; iv is wrong.

**14. Evaluate the statements related to Antibiotic Resistance:**

i. The use of antibiotics induces mutations in bacteria.

ii. Bacteria that acquire resistance through mutation survive and reproduce.

iii. Antibiotic resistance is an example of Natural Selection occurring within a short period.

iv. The multiplication of resistant bacteria is a major challenge in the field of medicine.

A) i and ii are correct; iii and iv are wrong B) ii, iii, iv are correct; i is wrong

C) i, ii, iii are correct; iv is wrong D) All statements are correct

**15. Evaluate the statements regarding Biological Evidence:**

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i. Organs that are similar in internal structure but different in function are called Homologous organs.

ii. Homologous organs indicate a common ancestral relationship.

iii. The human hand and a bird's wing are examples of homologous organs.

iv. Organs that are similar in function but different in internal structure are called homologous organs.

A) i, ii, iii are correct; iv is wrong B) ii and iv are correct; i and iii are wrong

C) i and iv are correct; ii and iii are wrong D) iii and iv are correct; i and ii are wrong

**16. Evaluate the statements related to Evolutionary Theories:**

i. According to Lamarck, unused organs eventually disappear over time.

ii. Darwin proposed the 'Theory of Natural Selection'.

iii. Organisms with favourable variations for survival are selected by nature.

iv. The inheritance of acquired characters is a major principle of Darwinism.

A) i and iii are correct; ii and iv are wrong B) ii, iii, iv are correct; i is wrong

C) i, ii, iv are correct; iii is wrong D) i, ii, iii are correct; iv is wrong

# Part 2: Assertion and Reason

**Question 1:**

Assertion (P): Superbugs are becoming a major threat to modern medicine.

Reason (R): Existing medicines are not effective against such bacteria that have acquired resistance to multiple antibiotics through mutation.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 2:**

Assertion (P): The human hand and a whale's flipper are homologous organs.

Reason (R): Although their internal structure is similar, they show differences in external appearance and function.

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A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 3:**

Assertion (P): RNA viruses evolve much faster compared to DNA viruses.

Reason (R): The probability of mutations occurring in RNA is higher compared to DNA.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 4:**

Assertion (P): Charles Darwin had a clear understanding of the genetic basis of the Theory of Natural Selection.

Reason (R): Gregor Mendel's laws of inheritance and knowledge about mutations were popular during Darwin's time.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 5:**

Assertion (P): Molecular biology provides strong evidence for the study of evolution.

Reason (R): Relationships can be found by comparing the sequence of nucleotides in DNA and the sequence of amino acids in proteins of different species.

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A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 6:**

Assertion (P): Humans and chimpanzees have a close evolutionary relationship.

Reason (R): There is no change in the number and sequence of amino acids in the beta chain of hemoglobin in humans and chimpanzees.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 7:**

Assertion (P): The struggle for existence among organisms paves the way for evolution.

Reason (R): Since resources are limited, only organisms with variations favorable for survival exist, and others perish.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 8:**

Assertion (P): The theory of 'Use and Disuse' is not scientifically accepted today.

Reason (R): Characters acquired through physical activities do not cause changes in genes and are therefore not passed on to the next generation.

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A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 9:**

Assertion (P): Fossils provide crucial evidence in the study of evolution.

Reason (R): Fossils are the remains or impressions of organisms that lived in ancient times.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 10:**

Assertion (P): The emergence of new variations in species is essential for evolution.

Reason (R): Without variations, new species cannot be formed through natural selection.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

**Question 11:**

Assertion (P): The human hand and a cat's forelimb are homologous organs.

Reason (R): Both are formed with the same internal structure to perform the same function.

A) The assertion is correct, but the reason is wrong.

B) Both the assertion and the reason are wrong.

C) Both the assertion and the reason are correct, but the reason is not the correct explanation of the assertion.

D) Both the assertion and the reason are correct, and the reason is the correct explanation of the assertion.

E) The assertion is wrong, but the reason is correct.

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## Part 3: Descriptive and Analytical Questions

1. "A young man with tuberculosis initially responded to antibiotics. However, months later, the same antibiotics were ineffective, and he died."
    a) Analyse this based on Darwin’s 'Survival of the Fittest'.
    b) Explain how mutation in bacteria caused this situation.
2. Giraffes' neck elongation is explained differently by Lamarck and Darwin.
    a) Compare Lamarckism and Darwinism using this example.
    b) How did the finding "Acquired variations do not change genetic structure" affect Lamarck’s theory?
3. Study the table showing amino acid differences in the Globin molecule between humans and various organisms.

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| ചിമ്പാൻസി | 0 |
| ഗൊറില്ല | 1 |
| എലി | 31 |

a) Based on this, identify the organism with the closest evolutionary relationship to humans and state the reason.

b) Explain how Molecular Biology provides evidence for evolution.

4. "Antimicrobial Resistance (Superbugs) is a global health problem."

a) Based on evolutionary theories, prepare awareness suggestions to prevent the formation of superbugs.

b) How does Evolutionary Medicine help in modern treatments?

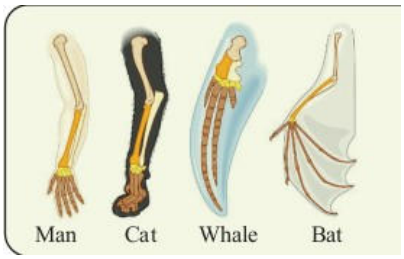
5. In drug users, reaction speed decreases and decision-making is delayed.

a) Which process in the nervous system is mainly affected?

b) What change occurs in the synapses?

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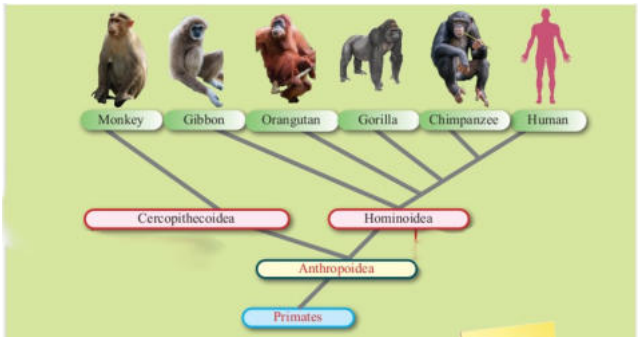
6. Observe the illustrations of the human hand, cat's forelimb, and whale's flipper.




a) What does the similarity in internal bone structure and difference in external form indicate?

b) How does this evidence support common ancestry?

7. Observe the human evolutionary tree.



a) What are the common features of the 'Hominoidea' group (humans and apes)?

b) How did the change in brain size influence human evolution?

8. "Darwin explained evolution through natural selection, but he couldn't explain how variations arise."

a) Which modern biological discoveries helped solve this limitation in Darwinism?

b) How did 'Neo-Darwinism' (Genetics + Darwinism) make evolutionary studies more logical?

9. During the Covid-19 period, different variants like Delta and Omicron emerged.

a) Why do RNA viruses evolve faster than DNA viruses?

b) Explain how mutations make viruses more dangerous, using textbook examples.

10. "Speciation is the formation of different species from a common ancestor."

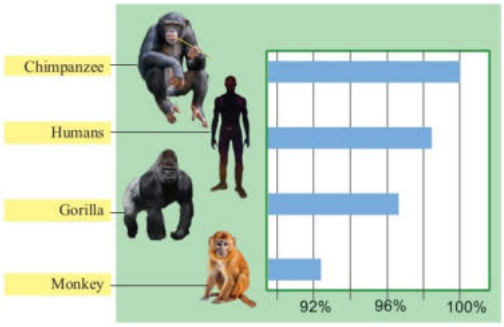
Compare the concepts of LUCA (Last Universal Common Ancestor) and MRCA (Most Recent Common Ancestor).

11. Fossils play a major role in proving evolution is a gradual process.

a) Why is *Archaeopteryx* called an 'intermediate link'?

b) What clues does it provide regarding the relationship between reptiles and birds?

12. Study the DNA sequence similarity between humans, chimpanzees, and gorillas.




a) How does an increase in DNA sequence similarity indicate evolutionary closeness?

b) Is genetic similarity more reliable than physical similarity in evolution? State your opinion.

13. In the human evolutionary tree, monkeys and apes are included in the Anthropoidea group.

a) What are the main differences between Cercopithecoidea (monkeys) and Hominoidea (apes and humans)?

b) Evaluate the importance of brain size progress from *Sahelanthropus tchadensis* to modern humans.

14. "DNA similarity between humans and chimpanzees is 98.8%, but it is much lower between humans and mice."

a) How can this be related to the concept of a 'Common Ancestor'?

b) How does molecular biology make evolutionary studies more accurate?

15. "Fossils are an open book of evolution, but the record is not always complete."

a) Why is *Archaeopteryx* called a 'Missing Link'?

b) What other modern methods are used to find the history of organisms without fossils?

16. "Viruses evolve fast, while human evolution takes millions of years."

a) Why is the mutation rate higher in RNA viruses (e.g., Covid)?

b) Why is viral evolution a threat to medicine? What precautions should we take?

**17.** "**A vital step in human evolution is the development of the Neocortex.**"

a) Evaluate the role of the Neocortex in intelligence, creativity, and language.

b) What would happen to human culture if the Neocortex hadn't developed?

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18. Analyse the statement and answer the questions.

"**A pesticide was used for years to kill mosquitoes. Initially effective, it later failed.**"

a) Relate this to Darwin's Natural Selection.

b) Will this change be passed to future generations? Why?

**19.** "**Genetic variations are the raw materials of evolution.**"

a) What is the role of mutation in causing variations?

b) How are beneficial variations preserved in natural selection?

20. A person cannot maintain balance while walking.

a) Which brain part is affected?

b) Write a major function of this part.

c) Write one consequence in daily life.

21. A person lost the ability to speak after an accident.

a) Which brain part is affected?

b) Write two functions of this part.

c) How does this make humans different from other organisms?

22. A person's heartbeat increases rapidly in an emergency.

a) Which nervous system is active?

b) Why is this response necessary?

23. A child withdraws their hand immediately after touching a hot vessel.

a) Name this action.

b) The center that controls this response?

c) Its significance?

24. Reaction speed decreases with drug use.

a) Which nervous system process is affected?

b) Change in synapses?

c) Danger involved?

d) A way to protect the nervous system.

25. Hand withdrawal after touching a hot object.

a) Name of the action.

b) Central nervous system part involved.

c) How does it help save life?

26. A person partially lost speech in an accident.

a) Which brain part is affected?

b) Two functions of this part.

c) Importance of the Neocortex.

27. A person responds bravely and quickly in an emergency.

Which nervous system is active?

Two functions of this system.

System that works opposite to this.

Combined role of these two systems.

28. Spinal cord was injured in an accident.

a) A major function of the spinal cord.

b) Which actions will be affected?

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c) Change in reflex actions?

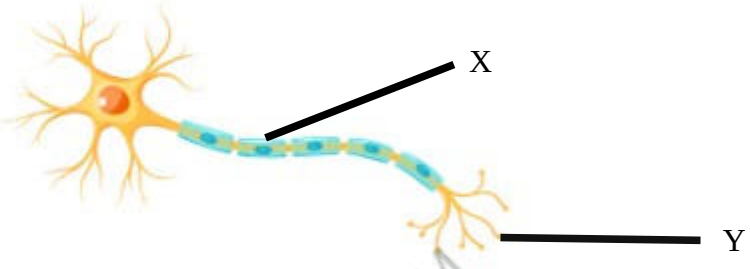
29.. Myelin sheath destruction is diagnosed in a person and the speed of response is slow.

a) One function of myelin sheath.

b) Change in impulse conduction.

c) Why is this dangerous?

30.. Analyse the diagram and answer the questions.




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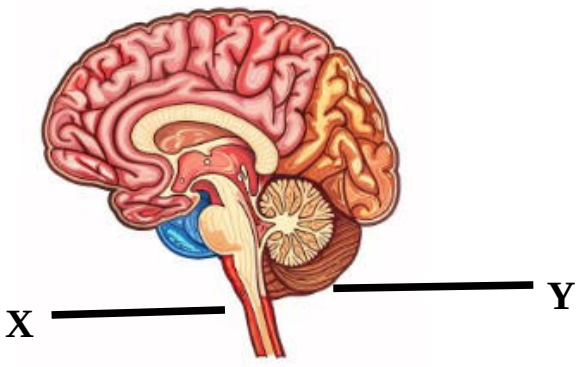
A. Redraw the diagram.

B. Identify the parts labelled **X** and **Y.**

C. Name the cells that produce myelin sheath in the nerves.

D. Predict what will happen if the part labelled **Y** becomes functionless?

31.. Evaluate the diagram and answer the questions.




A. Identify the parts labelled X and Y.

B. A drunk person lost his body balance. Which part between X and Y is affected by alcohol that caused the body imbalance of the person?

C. A patient was given pain killer after an injury. Name the part of the brain where pain killers act on?

D. What provides oxygen and nourishment to the brain tissues?

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# *Answer Key*

## Part 1: Multiple Choice Questions

**1. Statements about Lamarckism**

- **Correct Answer:** A) i, ii, iv are correct; iii is wrong.
- **Explanation:** The idea that giraffes' necks became long through natural selection belongs to Darwinism. According to Lamarck, the neck length increased due to continuous use.

**2. Statements about Antibiotic Resistance**

- **Correct Answer:** C) iii, iv are correct; i, ii are wrong.
- **Explanation:** Since antibiotic resistance is a genetic change (Mutation), it can be inherited. It is not an acquired character but a genetic variation.

**3. Statements about Superbugs**

- **Correct Answer:** D) i, iii, iv are correct; ii is wrong.
- **Explanation:** Superbugs, which are capable of resisting multiple antibiotics, cannot be easily destroyed by ordinary medicines.

**4. Statements about Darwinism**

- **Correct Answer:** A) i, ii, iv are correct; iii is wrong.
- **Explanation:** 'Inheritance of acquired characters' is Lamarck's theory, not Darwin's.

**5. Statements about the Evolution of Giraffes**

- **Correct Answer:** C) i, ii, iv are correct; iii is wrong.
- **Explanation:** Those with short necks failed in the struggle for existence; they did not undergo increased multiplication.

**6. Statements about Galapagos Finches**

- **Correct Answer:** C) ii, iii, iv are correct; i is wrong.
- **Explanation:** The finches have different types of beaks (i is wrong). Their evolution into different species is a process called speciation.

**7. Statements about Natural Selection**

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- **Correct Answer:** D) i, ii, iv are correct; iii is wrong.
- **Explanation:** Not all offspring survive. Only those with changes (variations) suitable for the environment survive.

**8. Speciation / Formation of New Species**

- **Correct Answer:** B) i, iii, iv are correct; ii is wrong.
- **Explanation:** Speciation is a very slow (Gradual) process, not something that happens suddenly.

**9. Statements about Fossils**

- **Correct Answer:** A) i, ii, iii are correct; iv is wrong.
- **Explanation:** Fossils are remains of ancient organisms and are not found in living organisms.

**10. Statements about Molecular Evidence**

- **Correct Answer:** C) ii, iii, iv are correct; i is wrong.
- **Explanation:** Molecular evidence (DNA, amino acid studies) supports the theory of evolution.

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**11. Statements related to Lamarckism**

- **Correct Answer:** A) i, ii, iv are correct; iii is wrong.
- **Explanation:** Experiments (e.g., Weismann's tail-cutting experiment) proved that acquired variations do not cause changes in genetic structure.

**12. Statements about Darwinism**

- **Correct Answer:** B) i, ii, iv are correct; iii is wrong.
- **Explanation:** Not all variations are beneficial. Organisms with harmful variations perish.

**13. Molecular Biological Evidence**

- **Correct Answer:** D) i, ii, iii are correct; iv wrong.
- **Explanation:** Based on the given options, D is appropriate (Though the fact is that there is no difference in the beta chain of haemoglobin between humans and chimpanzees).

**14. Antibiotic Resistance**

- **Correct Answer:** B) ii, iii, iv are correct; i is wrong.

- **Explanation:** Antibiotics do not induce mutation; instead, natural selection selects those that already have resistance through existing mutations.

**15. Biological Evidence (Homologous Organs)**

- **Correct Answer:** A) i, ii, iii are correct; iv is wrong.
- **Explanation:** Organs that are similar in function but different in internal structure are called **Analogous organs**.

**16. Statements related to Evolutionary Theories**

- **Correct Answer:** D) i, ii, iii are correct; iv is wrong.
- **Explanation:** The inheritance of acquired characters is Lamarck's principle, not Darwin's.

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## Part 2: Assertion and Reason

**Question 1:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation of the Assertion.
- **Explanation:** Superbugs are bacteria resistant to antibiotics. Their ability to resist multiple drugs through mutation is a challenge to modern medicine.

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**Question 2:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Organs with similar internal structure but different functions are called homologous organs. The human hand and a whale's flipper are examples.

**Question 3:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Mutations are more likely to occur in the structure of RNA compared to DNA. Therefore, RNA viruses evolve rapidly.

**Question 4:**

- **Answer:** B) Both Assertion and Reason are wrong.
- **Explanation:** Genetics had not developed during Darwin's time. He was unaware of Mendel's experiments or mutations, which was a limitation of Darwinism.

**Question 5:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Comparing the structure of DNA and protein helps in accurately understanding the evolutionary distance between organisms.

**Question 6:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** The lack of change in the sequence of amino acids in hemoglobin is evidence that humans and chimpanzees share a common ancestor.

**Question 7:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Since resources are limited, there is competition among organisms (Struggle for existence). The survival of those with favorable variations leads to evolution.

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**Question 8:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Lamarck's 'Use and Disuse' theory was proven wrong because acquired characters do not cause genetic changes and thus are not inherited.

**Question 9:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Fossils provide direct evidence of past life forms and how they evolved.

**Question 10:**

- **Answer:** D) Both Assertion and Reason are correct, and the Reason is the correct explanation.
- **Explanation:** Variations are the raw materials for natural selection. Without variations, adaptive changes to the environment would not occur.

**Question 11:**

- **Answer:** A) Assertion is correct, but the Reason is wrong.
- **Explanation:** The human hand and a cat's forelimb are homologous. However, the reason states they 'perform the same function', which is wrong as homologous organs usually perform different functions.

## Part 3: Descriptive and Situation-Based Answers

**1. Tuberculosis**

a) **Survival of the Fittest:** According to Darwin, those with favorable variations survive in the struggle for existence. Here, bacteria resistant to antibiotics (those with favorable variation) survived and multiplied, leading to the patient's death.

b) **Role of Mutation:** A sudden change in the bacteria's DNA (Mutation) gave them the ability to resist antibiotics.

**2. Lamarckism and Darwinism**

a) **Comparison:** Lamarck said giraffes got long necks by stretching for food, while Darwin explained that among giraffes with varying neck lengths, those with longer necks survived.

b) **Impact on the Theory:** The discovery that "acquired characters are not genetic" was a setback for Lamarckism. Physical changes like muscle strength cannot be inherited.

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**3. Molecular Biology**

a) **Closely Related Organism:** Chimpanzee.

**Reason:** There is zero difference in the sequence of amino acids in the beta chain of hemoglobin compared to humans.

b) **Evidence:** Similarity in DNA and protein indicates evolutionary closeness. Fewer differences mean a closer relationship.

**4. Superbugs and Evolutionary Medicine**

a) **Awareness Suggestions:**

- Complete the full course of antibiotics prescribed by the doctor.
- Avoid unnecessary use of antibiotics.

b) **Role of Evolutionary Studies:** Understanding how pathogens evolve to resist drugs helps in developing new treatments and medicines.

**5. Drugs and Social Life**

a) Functioning of neurotransmitters is affected.

b) Flow of impulses in synaptic gaps is obstructed.

c) Leads to behavioral changes, aggression, and breakdown of family relationships.

**6. Homologous Organs and Evolution**

a) **Indication:** Similarity in the internal bone structure of a human hand, cat's forelimb, and whale's flipper proves they had a **common ancestor**. External differences are adaptations to different environments and uses (grasping, walking, swimming).

b) **Common Ancestry:** These similarities support the idea that these organisms originated from the same lineage and evolved over time under different conditions.

**7. Hominoidea**

a) **Common Features:** They are tailless, have large brains, protect offspring for long periods, have mobile limbs, and developed social relationships.

b) **Influence of Brain Size:** Increased brain size and complexity helped humans make tools, use fire, develop language for communication, and survive complex situations.

**8. Neo-Darwinism**

a) **Findings that helped:** Discoveries in genetics like mutation, genetic recombination, and gene flow helped explain how variations occur.

b) **Logic:** By combining Darwin's natural selection with genetics, Neo-Darwinism provided a precise scientific explanation of how evolution happens.

**9. Viruses and Evolution**

a) **Evolution of RNA Viruses:** RNA viruses lack 'proofreading' ability when replicating their genetic structure. Thus, mutations occur rapidly, leading to new variants.

b) **Danger:** Mutations change surface proteins (like spike proteins), helping the virus evade our immune system and vaccines.

**10. Speciation**

a) **LUCA vs MRCA: LUCA** (Last Universal Common Ancestor) is the very first ancestor common to all life on Earth. **MRCA** (Most Recent Common Ancestor) is the closest ancestor shared by a specific group (e.g., humans and chimps).

b) **Process: Geographical isolation** prevents members of a species from interbreeding. Over time, different mutations and variations accumulate, leading to **reproductive isolation** and the formation of a new species.

**11. Archaeopteryx**

a) **Missing Link:** It is called a missing link because it possesses both reptilian features (teeth in jaws, vertebrae in tail) and avian features (feathers, wings).

b) **Indication:** It provides clear evidence that birds evolved from reptiles.

**12. DNA Matching**

a) **Indication:** Higher percentage of DNA sequence matching shows closer evolutionary relationship. It implies a common ancestor and a more recent divergence.

b) **Opinion:** Genetic similarity is more reliable because physical structures can change due to the environment (Analogous organs), but DNA accurately records evolutionary history.

**13. Anthropoidea**

a) **Difference:** Cercopithecoidea (monkeys) have tails and smaller brains. Hominoidea (apes and humans) are tailless and have larger brains.

b) **Brain Progress:** The increase in **Cranial capacity** from *Sahelanthropus* to modern humans was crucial for intellectual growth, social progress, and complex lifestyles.

**14. Role of Molecular Biology**

a) **Relationship:** High DNA similarity between humans and chimps proves a recent common ancestor. Lower similarity with mice shows divergence much earlier in the evolutionary tree.

b) **Accuracy:** Molecular biology makes evolutionary studies objective by mathematically calculating differences in amino acids and nucleotides.

**15. Fossils and Evolution**

a) **Archaeopteryx - Missing Link:** Possession of both reptilian and avian traits proves evolution from reptiles to birds.

b) **Other Methods:** Modern science uses molecular biology (DNA/protein comparison), comparative anatomy, and embryology.

**16. Speciation Flowchart**

a) Common Ancestor -> Geographical Isolation -> Different Environments -> Different Variations -> Natural Selection -> Reproductive Isolation -> New Species.

b) **Role:** Mutation creates new genetic variations. Natural selection selects and preserves those variations that are suitable for the environment.

**17. Brain Development and Neocortex**

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a) **Difference:** Development of the **Neocortex** enabled humans for high-level thinking and logical decisions.

b) **Brain Size and Tools:** As the brain evolved, intelligence to make complex tools was acquired. Neocortex is the centre for intelligence, creativity, and language. Without it, humans would have remained mere animals.

**18. Pesticide Resistance - Darwinism**

a) **Natural Selection:** Mosquitoes with the ability to survive the pesticide (favorable variation) survived and multiplied.

b) **Inheritance:** Yes, since it is a genetic change (Mutation), it is passed to future generations.

**19. Genetic Variations**

a) **Mutation:** Sudden changes in genes provide new variations for evolution.

b) **Protection:** Beneficial variations are preserved as those organisms survive and pass traits to offspring.

**20. Inability to maintain balance while walking:**

a) **Part: Cerebellum**.

b) **Function:** Coordinates muscular activities and maintains body balance.

c) **Effect:** Cannot perform precise movements, prone to falling.

**21. Loss of speech after an accident:**

a) **Part: Cerebrum**.

b) **Function:** Center of intelligence, thought, and senses; analyzes messages.

c) **Reason:** Developed cerebrum allows humans to use language and analyze complex things.

**22. Increased heartbeat in emergency:**

a) **System: Sympathetic Nervous System**.

b) **Reason:** Rapid blood flow is necessary to deliver more oxygen and energy to muscles to face the situation.

**23. Withdrawing hand immediately after touching a hot vessel:**

a) **Action: Reflex action**.

b) **Center: Spinal cord**.

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c) **Significance:** Responding quickly without wasting time to think protects the body from severe burns.

**24. Drug use and response speed:**

a) **Process: Impulse transmission** at synapses.

b) **Mechanism:** Affects production and function of **Neurotransmitters**.

c) **Danger:** Inability to respond in time during emergencies can be fatal.

d) Advice: Avoid drugs, maintain exercise and sleep.

**25. Reflex action**

a) **Process: Reflex action**.

b) **Center: Spinal cord**.

c) **Significance:** Ensures safety by reducing response time.

**26. Speech ability**

a) **Part: Cerebrum**.

b) **Function:** Controls higher mental functions; analyzes messages.

c) **Significance:** Makes humans superior in imagination and creativity.

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**27. Response in emergency:**

a) **Sympathetic System:**

b) Increases heartbeat, dilates pupil.

c) **Parasympathetic System:** Returns body to normal.

d) **Significance:** Both work together for **Antagonistic action** to maintain internal stability.

**28. Spinal cord injury**

a) **Function:** Message transmission.

b) **Effect:** Motor pathways are blocked

c) reflexes are partially or fully lost.

**29. Myelin sheath destruction**

a) **Function:** Protect nerve fibres.

b) **Effect:** Impulses go astray or slow down

c) coordination of nervous system fails.

**30. A. Diagram.**

**B. X- Myelin sheath.**

**Y- Synaptic knob.**

**C. Shwann cells.**

**D. The impulse transmission through the synapsis will be affected.**

**31. A. X-Medulla oblongata.**

**Y- Cerebellum .**

**B. Y, cerebellum .**

**C. Thalamus.**

**D. Cerebro spinal fluid.**

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# unit 3

## Behind sensations.

**MULTIPLE CHOICE**

*Choose the most scientifically accurate option.*

1. A person moves quickly from a sunny garden into a dark storage room. Initially, they cannot see anything, but after a few minutes, they begin to perceive the shapes of objects. This "dark adaptation" is primarily due to:

a) The rapid contraction of circular muscles in the iris.

b) The regeneration of rhodopsin in rod cells which was bleached in bright light.

c) The immediate activation of cone cells to detect low-intensity light.

d) The vitreous humour changing its density to allow more light to the yellow spot.

2. If a gymnast performs a rapid series of somersaults and stops, they feel dizzy for several seconds. This occurs because:

a) The cochlea is still vibrating from the sound of the movements.

b) The endolymph in the semicircular canals continues to move, stimulating hair cells.

c) The auditory nerve is temporarily overloaded with balance impulses.

d) The Eustachian tube has failed to balance the internal ear pressure.

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3. You accidentally touch a hot vessel and immediately withdraw your hand. Which of the following correctly traces the flow of information for this **nociceptive (pain)** response?

a) Brain → Sensory Neuron → Nociceptor → Muscle.

b) Stimulus → Nociceptor → Spinal Nerve → Brain → Motor Response.

c) Receptor potential → Muscle → Action potential → Optic Nerve.

d) Chemoreceptor → Olfactory Nerve → Brain → Glandular Response.

**(Assertion and Reason)**

***For each question, choose the correct option from the following*:**

**A**: Both Assertion (A) and Reason (R) are true, and (R) is the correct explanation of (A) .

**B**: Both Assertion (A) and Reason (R) are true, but (R) is **not** the correct explanation of (A) .

**C**: Assertion (A) is true, but Reason (R) is false.

**D**: Assertion (A) is false, but Reason (R) is true.

**4.Assertion (A):** A person immediately withdraws their hand upon touching a hot vessel.

**Reason (R):** Nociceptors(pain) in the skin detect the heat and transmit impulses through sensory neurons to the brain to create a response.

**5.Assertion (A):** Vitamin A deficiency can lead to night blindness.

**Reason (R):** Retinal, a component of the light-sensitive pigment rhodopsin in rod cells, is formed from Vitamin A.

**6.Assertion (A):** The pupil constricts when a person enters a brightly lit room.

**Reason (R):** The radial muscles of the iris contract to reduce the amount of light entering the eye to protect the retina.

**7**.**Assertion (A):** Swallowing or yawning helps relieve ear discomfort during a flight's take off.

**Reason (R):** The Eustachian tube connects the middle ear to the pharynx and helps balance the air pressure on both sides of the tympanum.

**8**.**Assertion (A):** Dizziness is experienced for a short time after someone stops spinning in circles.

**Reason (R):** The endolymph in the semicircular canals continues to move due to inertia, stimulating hair cells even after physical motion has stopped.

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**9**.**Assertion (A):** Snakes flick their tongues frequently to "smell" their surroundings.

**Reason (R):** The tongue carries chemical particles to the Jacobson's organ located in the roof of the mouth for analysis.

## SHORT ANSWER

**10**.A patient has a blockage in the duct that reabsorbs **aqueous humour** into the blood. Explain the physiological consequence of this blockage on the optic nerve and the resulting vision disorder.

**11**.How does the structure of the **middle ear** (specifically the ear ossicles) ensure that vibrations from the large **tympanum** are effectively transferred to the much smaller **oval window**?

**12**. **Evolutionary Biology:** Compare the **Jacobson's organ** in a snake to the **olfactory receptors** in a human. Why is the snake's method of "flicking" its tongue an essential part of its sensory experience compared to human breathing?

Long answer type

**13**.A child is unable to distinguish between red and green colors but can see blue clearly.

- Identify which specific photoreceptors are likely malfunctioning.
- Using your knowledge of genetics, explain why this condition is more frequently observed in males than females.

**14.Pressure Regulation:** During a rapid descent in an air plane, passengers are often advised to swallow or chew gum.

- Which structure connects the middle ear to the pharynx to allow this?
- Explain how this action prevents the **tympanum** from rupturing due to atmospheric pressure changes.

**Very long answer type**

**15**. Define the process of **binocular vision**.

- Evaluate how the 3D vision resulting from this fusion provides a survival advantage for a predator (like a hawk) compared to an organism with eyes on the sides of its head.

**16:** You smell a delicious meal (smell) and your mouth begins to water (salivation).

a. Identify the types of receptors stimulate.

b. Describe the role of the **brain** in analysing these impulses and sending instructions to the **muscles/glands** to create this specific response**.**

17. Analyse the physiological necessity of the continuous production and reabsorption of aqueous humor. What would be the systemic consequence if this cycle were disrupted, specifically regarding the optic nerve?

18. Compare and contrast the roles of glutamate in the dark and in the presence of light within the retina. How do on bipolar cells and off bipolar cells utilize this single neurotransmitter to differentiate between vision and darkness?

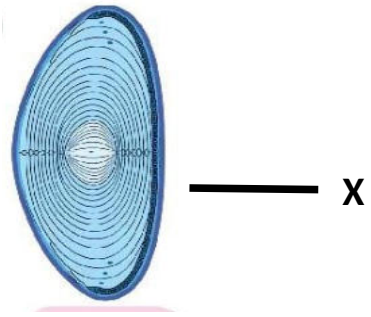
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19. Evaluate the structural differences between compound eyes in insects and high-vision eyes in hawks. How do these diverse sensory mechanisms reflect the survival requirements of their respective ecological niches?

20.Explain the mechanical advantage provided by the size difference between the tympanum and the oval window. How does the Eustachian tube prevent these delicate mechanics from failing during rapid changes in atmospheric pressure?

21. Describe how the brain integrates information from the vestibular system, eyes, and muscles to maintain body balance. Why is the rotational movement of the head specifically detected by the semicircular canals rather than the vestibule?

22. Identify the given structure and answer the questions.

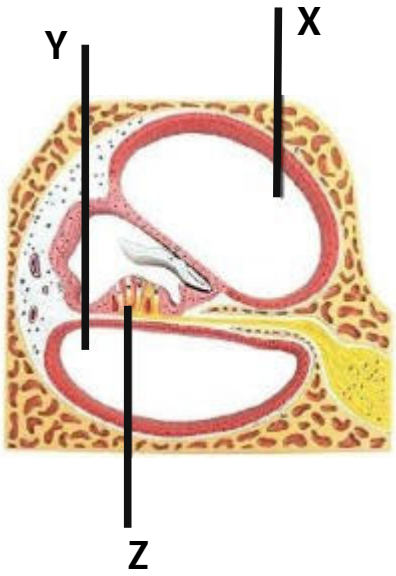


a. Name the Structure X

b. What is the main structural component X.

c. How does age related changes affect this structure?'

23. Observe the diagram and answer the questions.




a. Name X and Y.

b. Name the fluid present in X and Y

Identify the part Z and mention its role .

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## Answer key :

1. **Answer: (b)** The regeneration of rhodopsin in rod cells which was bleached in bright light.

*Reasoning:* In bright light, rhodopsin (found in rod cells) is split. When moving to a dark room, rod cells must regenerate rhodopsin from opsin and retinal (Vitamin A) to detect low-intensity light, a process that takes several minutes.

**2.Answer: (b)** The endolymph in the semicircular canals continues to move, stimulating hair cells.

*Reasoning:* Semicircular canals are perpendicular to each other and contain endolymph. When the head stops rotating, the fluid continues to move due to inertia, stimulating hair cells and sending false "movement" signals to the brain.

**3.Answer: (b)** Stimulus – Nociceptor(pain) -Spinal Nerve -Brain -Motor Response.

*Reasoning:* The process of identifying pain (nociception) involves receptors called nocireceptors transmitting impulses through sensory neurons to the spinal nerve and then to the brain.

**4.Answer: A:** Touching a hot vessel stimulates nociceptors (pain receptors). The resulting impulse travels via sensory neurons to the brain, which then instructs the muscles to withdraw the hand.

**5.Answer: A:** Rhodopsin is essential for vision in dim light. Since retinal is derived from Vitamin A, a deficiency directly prevents the formation of enough rhodopsin, causing night blindness.

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**6.Answer: C:** The Assertion is true (pupil constricts in bright light), but the Reason is false because in bright light, **circular muscles** contract to constrict the pupil, not radial muscles (radial muscles contract in dim light to dilate the pupil).

**7.Answer: A :** Swallowing opens the Eustachian tube, allowing air to pass between the pharynx and middle ear, thereby equalizing internal and external pressure on the eardrum.

**8.Answer: A :** The semicircular canals detect rotational movement. When you stop spinning, the fluid (endolymph) inside continues to move briefly, sending signals to the brain that the body is still in motion, which creates the sensation of dizziness.

**9.Answer: A :** Snakes lack a traditional nose for smelling like humans; instead, the Jacobson's organ is their primary chemoreceptor for the air, and the tongue is the transport mechanism for those chemical stimuli.

**10.Glaucoma Diagnosis: Mechanism:** Aqueous humor is normally oozed from blood-like tissue and reabsorbed back into the blood to regulate eye pressure.

1. **Consequence:** If reabsorption fails, pressure within the aqueous chamber increases.
2. **Outcome:** This high pressure eventually damages the **optic nerve**, leading to permanent vision loss (Glaucoma).

**11.Acoustic Physics (Middle Ear): Mechanism:** The **ear ossicles** (malleus, incus, and stapes) act as a bridge.

3. **Transfer:** They pick up vibrations from the larger **tympanum** and concentrate that energy onto the much smaller **oval window**.
4. **Advantage:** This size difference and the lever action of the bones amplify the sound vibrations so they can effectively move the fluids (perilymph/endolymph) in the inner ear.

**12.Evolutionary Biology (Snake vs. Human Smell):** * **Snake:** Snakes use the **Jacobson's organ** located in the roof of the mouth. They flick their tongue to collect chemical particles from the air and transfer them to this organ for analysis.

5. **Human:** Humans use **olfactory receptor neurons** in the nasal mucus membrane. Particles are inhaled and dissolved in mucus to stimulate these neurons.
6. **Survival:** Snakes rely on this for tracking prey and navigation in environments where vision might be limited.

**13.Colour Vision: Malfunctioning Receptors:** The **M-cones** (green light sensitivity) and **L-cones** (red light sensitivity).

7. **Genetic Reason:** The genes for red and green cone pigments are located on the **X chromosome**. Since males have only one X chromosome, a single defective gene results in the disorder, whereas females need two defective genes (one on each X) to be affected.

**14.Pressure Regulation:** * **Structure:** The **Eustachian tube/canal**.

8. **Mechanism:** It connects the middle ear to the pharynx. Swallowing opens this tube, allowing air to move in or out of the middle ear to equalize pressure with the atmospheric air in the outer ear, protecting the **tympanum** from stretching or rupturing.

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**15.Binocular Vision & Survival: :** The process where the brain receives two slightly different images from each eye (due to different angles) and combines them into one.

9. **Predatory Advantage:** Fusion results in **3D vision**, which is critical for perceiving **depth** and determining exact distances. This allows a predator like a hawk to strike prey with high precision.

**16.Sensory Integration:** * **Receptors: Chemoreceptors** (in taste buds) and **Olfactory receptor neurons** (in the nose).

10. **Neural Process:** 1. **Stimulus:** Chemical molecules dissolve in mucus/saliva.

    2. **Impulse:** Receptors generate electrical impulses.

    3. **Brain Analysis:** Impulses reach the sensory centres of the brain where they are analysed and recognized as "food".

    4. **Motor Instruction:** The brain sends instructions to the **salivary glands** to produce saliva as a physiological response.

……

Profound level

**17. Answer: Aqueous Humor and Optic Health**

8. **Physiological Cycle:** Aqueous humor is a watery fluid that oozes from blood-like tissue fluid and is constantly reabsorbed back into the blood. This cycle provides nutrients and oxygen to the **lens and cornea**, which lack their own blood supply.
9. **Systemic Consequence:** If reabsorption fails, internal pressure within the aqueous chamber increases. This pressure is transmitted to the back of the eye, eventually damaging the **optic nerve**, leading to a condition known as **glaucoma** and potential blindness.

**18. Answer: Glutamate and Bipolar Cell Activation**

10. **In Darkness:** Photoreceptors continuously produce **glutamate**. This activates **off bipolar cells** (sensing darkness) and inactivates **on bipolar cells**, sending "darkness" impulses to the brain.
11. **In Light:** When exposed to light, photoreceptors stop producing glutamate. This absence of glutamate causes **on bipolar cells** to become active while **off bipolar cells** become inactive, creating the sense of vision.

**19. Answer: Comparative Visual Adaptations**

**Insects:** Possess **compound eyes** made of thousands of **Ommatidia**. This structure is ideal for detecting fast motion from multiple angles, essential for flight and avoiding predators.
**Hawks:** Have highly developed eyes capable of long-distance vision and detecting **ultraviolet radiation**. This adaptation allows them to track small prey from extreme heights, providing a specialized hunting advantage.

**20. Answer: Pressure Balance and Acoustic Amplification**

- **Acoustic Advantage:** The **tympanum** is much larger (9-10 mm diameter) than the **oval window**. This allows sound vibrations to be concentrated and amplified as they pass through the **ear ossicles**, effectively moving the denser fluids in the cochlea.
- **Pressure Regulation:** The **Eustachian tube** connects the middle ear to the pharynx. By opening during swallowing or yawning, it balances the air pressure on both sides of the tympanum, ensuring it remains flexible enough to vibrate properly.

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**21. Answer: Neural Integration for Balance**

- **Integration:** The brain maintains balance by simultaneously receiving and analyzing impulses from the **vestibular nerve** (inner ear), **optic nerve** (visual orientation), and **muscles** (proprioception).
- **Rotational Detection:** The **semicircular canals** are arranged perpendicular to each other and contain **endolymph**. When the head rotates, the inertia of the fluid stimulates hair cells within the canals, specifically signalling the direction and speed of the turn to the brain.

22. a. Lens

b. Crystalline a protein.

c. Age related changes affect the flexibility and transparency of lens as well as vision.

23. a. Upper chamber,Lower chamber.

b. Perilymph

c. Organ of Corti. It contains the auditory receptors which produce impulse when vibrations reach there.

# Unit:4

# Chemoreception in Organisms.

## ONE MARK QUESTIONS

Analyse the given **Assertion** and **Reason** and choose the correct answer.

1.Assertion : Plants tip grows towards sun light
Reason : Growth of plant tip is due to the action of Auxin

a) Assertion is true, but reason is false
b) Assertion is false , but reason is true
c) Both assertion and reason are true
d) Both assertion and reason are false

2.Assertion : Phytochrome is synthesized in fully developed leaves
Reason : Phytochrome has a role in capturing energy for glucose production.

a) Assertion is false but reason is true
b) Both assertion and reasons are true
c) Assertion is true but reason is false
d) Both assertion and reason are false

3 Assertion : Para follicular cells in thyroid gland produce thyroxine
reason : Thyroxine controls metabolism and body temperature
a) Assertion true but reason is false
b) Assertion false but reason is true.
c) Both assertion and reason are false
d) Both assertion and reason are true

4 Assertion :Calcitonin is produced when the level of calcium in the blood increases.
Reason : Calcitonin regulates the calcium level by reducing the mixing of calcium from bones into the blood.

a)Both assertion and reason are correct
b)Assertion is correct and reason is false
c)Both assertion and reason are false
d)Assertion is false, reason is correct

5 Assertion: Pheromones are the chemical substance secreted into the surroundings by certain organism.
Reasons: Pheromones help to regulate metabolic activities in organisms.

a) Both the assertion and reasons are correct
b) Assertion is correct and reasons is false
c) Both assertion and reasons are false
d) Assertion is false and reasons are correct

6 Assertion : Pineal gland is known as biological clock.
Reason : Melatonin controlling the activities at regular intervals throughout the day.

a) Assertion is false and reasons are correct
b) Both assertion and reasons are correct
c) Assertion is correct and reason is false
d) Both assertion and reasons are false

7 Assertion: Auxins accumulate in the shaded part of the plant stem, which causes the stem to bend towards the light.
Reason : Auxins increase cell elongation in the cells where they accumulate.

a) Both assertion and reason are correct. The reason is the correct explanation of the statement.
b) Both assertion and reason are correct. The reason is an incorrect explanation of the statement.
c) Assertion is correct. Reason is incorrect.
d) Assertion is incorrect. Reason is correct

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8 Assertion :Cytokinins help in seed germination.
Reason : Cytokinins stimulate cell division and differentiation in seeds

a) Assertion is false, reason is correct
b) Both assertion and reason are false
c) Both assertion and reason are correct
d) Assertion is correct reason is false

9 Which of the following is NOT a function of the female sex hormone oestrogen?
a) Controlling the growth of female reproductive organs
b) Controlling secondary sexual characteristics
c) Maintaining the foetus in the uterus
d) Playing a role in the menstrual cycle

10 Analyse the statement and reason and find the correct one.

Statement :When the glucose level in the blood decreases, the alpha cells of the pancreas produce insulin.
Reason:The hormone insulin accelerates the entry of glucose in to the cell

a)Both statement and reason are incorrect
b)Statement is correct, reason is incorrect
c)Statement is incorrect, reason is correct
d)Both statement and reason are correct

11 Statement:Thymus gland helps in the maturation of T lymphocytes
Reasons:It plays major role in the body's immune system.

a)Both statement and reason are incorrect
b)Statement is correct, reason is incorrect
c)Statement is incorrect, reason is correct
d)Both statement and reason are correct

12 Analyse the following statements
related with the regulation of calcium in blood and find the correct answer

i)When the level of calcium increases in blood parathyroid gland produces calcitonin
ii)The normal level of calcium in blood is 9-11mg/dL
iii) When the level of calcium decreases in blood parathormone accelerate the reabsorption of calcium from kidney to blood
iv) Calcitonin and parathormone are produced by thyroid gland

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Options
a) i and ii are correct
b) ii and iv are correct
c) ii and iii are correct
d) i and iv are correct

13 The normal level of calcium is maintained by the combained action of two hormones .They are
a) Thyroxine and calcitonin
b) Thyroxine and parathormone
c) Parathormone and cortisol
d) Calcitonin and parathormone

14 Find out the correct option regarding the function of calcitonin

i) Stimulate the activity of mixing of calcium and phosphate from the bone into the blood.
ii) Decreases the absorption of calcium to blood
iii) Accelerates the absorption of calcium to blood

iv) Inhibits the activity of mixing of calcium and phosphate from the bone into the blood

options
a) i and ii are correct
b) ii and iv are correct
c) i and iii are correct
d) All are correct

15 Evaluate the statements related with the function of parathormone

i) Accelerates the reabsorption of calcium and phosphate from kidney to blood
ii)Decreases the absorption of calcium to blood
iii) Inhibits the activity of mixing of calcium and phosphate from the bones into the blood
iv) Stimulates the activity of mixing of calcium and phosphate from the bone into the blood

options
a) i,ii,iii correct,iv incorrect
b) i,iv correct,ii,iii incorrect
c) i correct, ii,iii,iv incorrect
d) i,iii correct , ii,iv incorrect

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16 Evaluate the statements and find correct answer

i) Gibberellins maintains the dormancy of seed
ii) Auxins promote the transport of nutrients to growing regions.
iii) Ethylene plays a role in the degradation of protein and nucleic acid
iv) Abscisic acid inhibits the growth of plant parts

a) i,ii,iii are correct iv incorrect
b) i,iii correct ii,iv incorrect
c) i,ii incorrect iii,iv correct
d) i,ii,correct iii,iv incorrect

17 Choose the correct option from the following

i) 2,4-D is a synthetic auxin
ii) Root formation is induced by synthetic gibberellins
iii) Synthetic ethylene is used in fruit ripening
iv) Auxin is used in increasing the size of fruit

a) i,ii is correct iii,iv incorrect
b) i,iii correct ii,iv incorrect

c) i,ii,iii,correct  iv incorrect
d)  i,ii,iii, incorrect iv correct

18  Find out the correct option regarding plant hormones

i) auxin stimulate the formation of flower and fruit
ii) gibberellins promotes the transport of nutrients to growing region
iii) cytokinins helps growth of fruits and seeds
iv) ethylene plays an important role in ageing of leaves and flowers

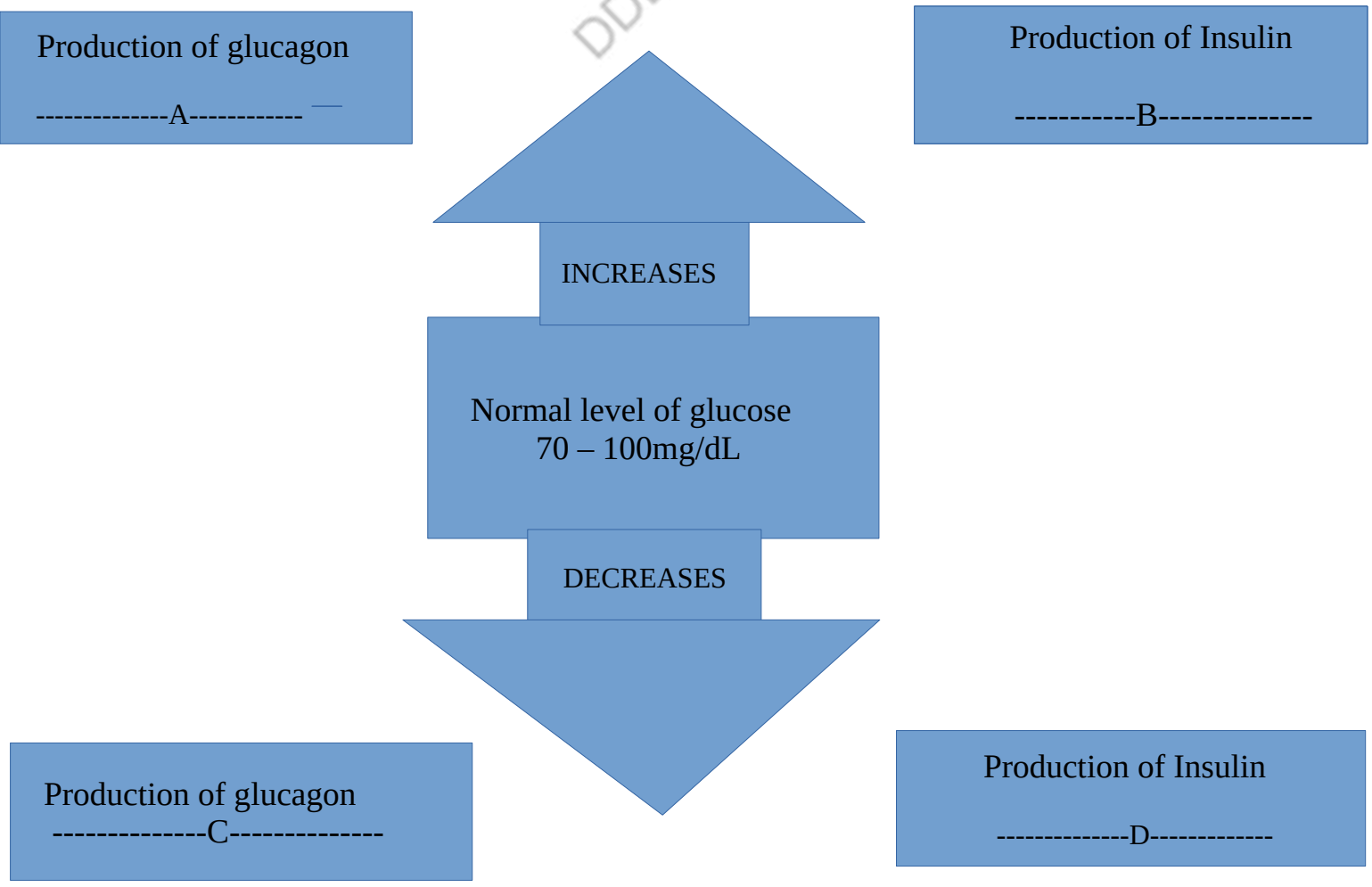
a) i,ii,iii correct  iv incorrect
b) i correct  ii,iii,iv incorrect
c) i,iv correct ii,iii incorrect
d) i incorrect ii,iii,iv correct

## Short Answer type Questions.

1  Analyse  and complete the illustration

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2 When a person sees a snake,he suddenly becomes alert and his heart rate increases.
a)Which hormone causes this response ?
b)How does it help in emergency situations ?

3. Information about a chemical substance is provided in the box. Analyse it and answer the questions.

- Organisms belonging to the same species secrete it into the sorroundings for communications
- It is used to mark a path and to attract mates.

a)Which chemical substance is mentioned in the box?
b)Write any two other properties of this chemical substance ?

4. During child birth ,contraction of uterine muscle occurs due to a hormone.
a) Name the hormone ?
b) Where is it produced ?

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5. If the amount of thyroxine in the blood decreases,through various steps ,the amount of thyroxine is brought back to an adequate level. List the actions that takes place for this in the correct order and mention the hormones involved in it.

6. A woman experiences irregular menstrual cycle and difficulty in maintaining pregnancy.

a)Identify the hormone involved ?
b) Name the gland that secretes it ?

7.A mother is unable to produce milk after delivery.

a) Name two hormones that help in the production of milk.
b) One among the two hormones help in the contraction of uterine muscles during child birth. Which lobe of pituitary gland secretes it?

8. A patient with long term uncontrolled diabetes develops blurred vision and numbness in feet.

Identify the conditions and explain their cause.

9. Observe the diagram and answer the questions.



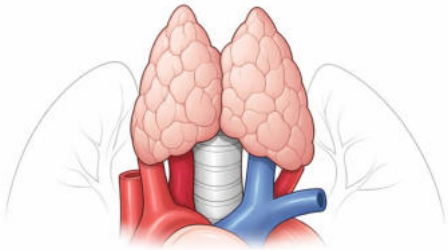
a) Identify the plant hormone responsible for the bending of the shoot.

b) write any other function of this hormone.

10. A person shows weight loss, increased heart beat and nervousness .
Identify the disorder and explain the reason behind these symptoms?

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11.Observe the diagram and answer the questions



A child has a congenital absence of the gland shown in the picture.

a) Identify the gland?

b) Predict how does the above condition affect the immune system of the child?

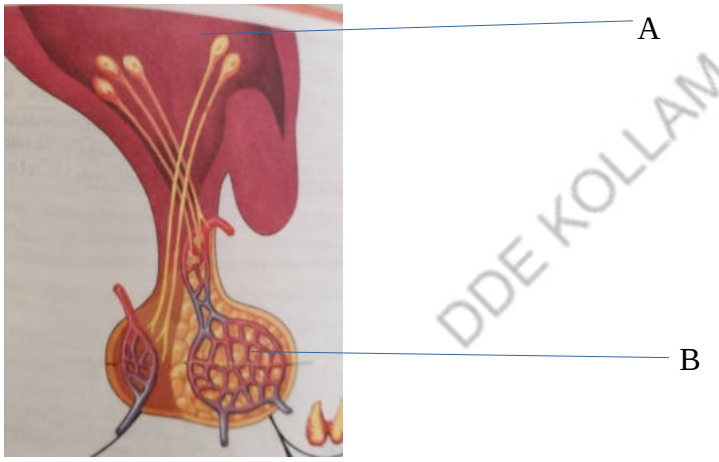
12. Using a mobile phone late at night causes sleeping problems."

a) Write the name of the hormone affected in this situation.
b) How does this activity affect sleep?
c) Explain the reason.

13. A gland, together with the sympathetic nervous system, prepare the body to overcome an emergency.

a) Identify the gland? Which are the hormones secreted by this gland
b) How do these hormones act in the body to deal with an emergency

14. Observe the diagram and answer the questions.




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a) Write the name of the gland labelled A in the diagram.

b) Write the names of any two hormones secreted by the gland labelled B.

c) Which gland is stimulated by TSH?

15. "Phytochrome plays an role in flowering " .Photoperiodism is the process helps in flowering of angiosperms

a) Are these statements correct ?
b) Write how are they related to each other ?

16. Read the following sentence and the questions.

Thyroxine is a major hormone that controls the metabolism and body temperature

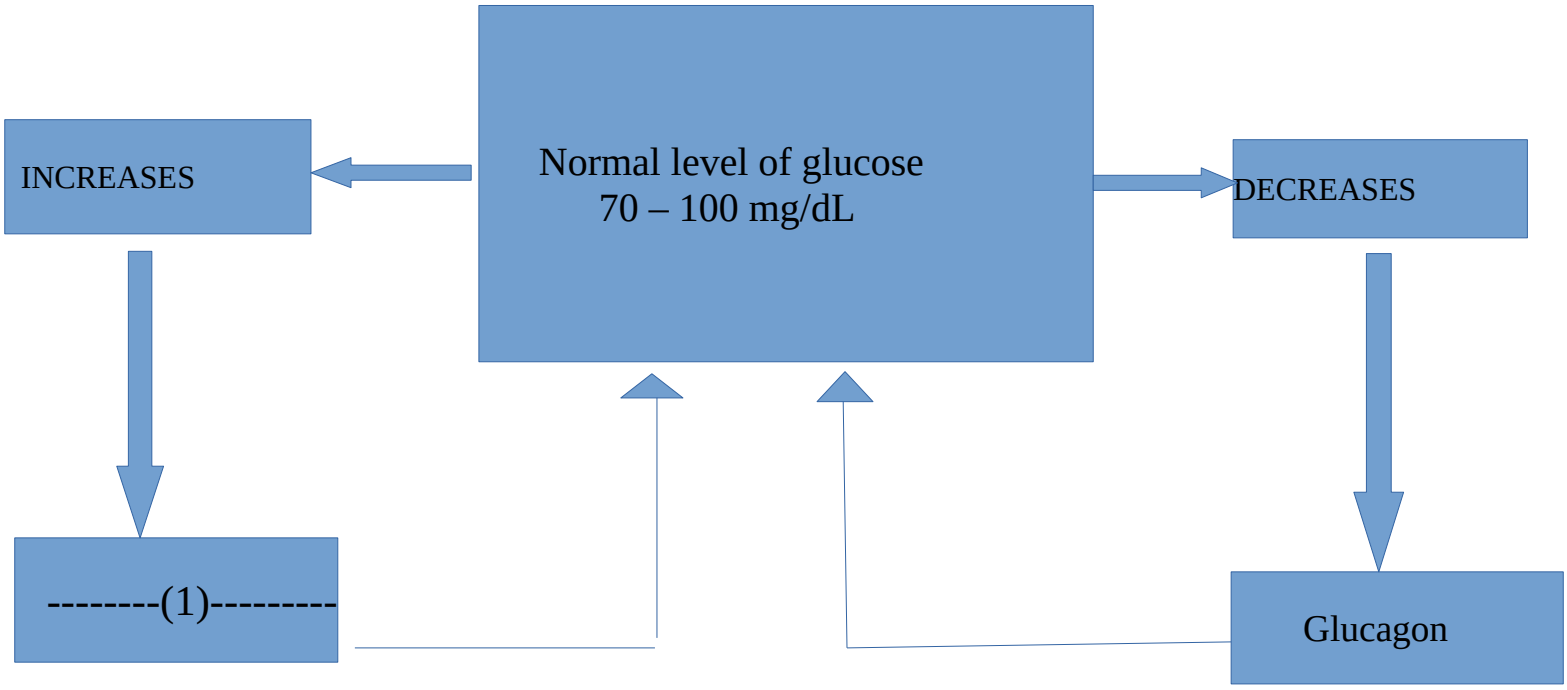
a) which part of which gland produce this hormone ?
b) write two other functions of this hormone ?

17 Explain how hormones influence plant growth in the following situations

a) When a farmer spray synthetic gibberellins in plants
b) Ethylene is widely used in agricultural sector

18. The illustration given below is related to the regulation of glucose in blood .
Observe the illustration and answer the question

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a) Which hormone is indicated as (1)

b) Redraw the diagram showing calcium instead of glucose.

**19. "Control diabetes – don't let it stand in your way".** This is a message shown in a hospital.

a) What will be the consequences if diabetes is not controlled ?
Mention any two.
b) What is the cause of Type 1 diabetes

20. Evaluate the indicators related to hormonal disorders and answer the questions.

- Related to growth hormones

- Occurs after the growth phase

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a) Name the hormonal disorder mentioned here ?

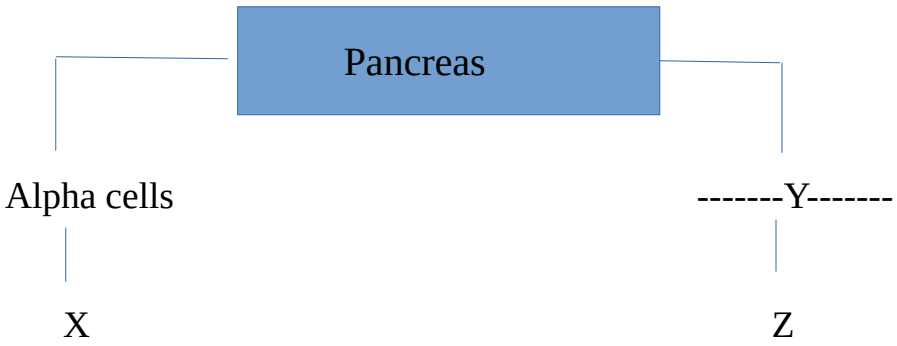
b) Write the cause and symptoms of this disorder.

21..Using the given hints, prepare a table with a suitable heading.

Hints :

Production of ADH increases
Production of ADH decreases
Kidneys increase the reabsorption of water
Kidneys decreases the reabsorption of water
Production of urine increases
Production of urine decreases

22. Analyse the illustration and answer the questions ?




a)Name the hormone X ?
b)How does the hormone Z regulate blood glucose level ?
c)What condition is diagnosed when fasting blood glucose rises above 126mg/dL ?
What are its symptoms?

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# ANSWER KEY

1. c
2. c
3. b
4. a
5. b
6. b
7. a
8. c
9. c
10 .c
11. d
12 . c
13 .d
14 .b
15 .b
16 . c
17 . b
18 .c

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**Short Answer type Questions.**

1. A – Decreases
   B – Increases
   C – Increases
   D – Decreases

2. a ) Epinephrine
   b) It prepare the body to deal with an emergency situation through a number of physiological changes such as an increase heart rate and blood pressure,increase blood glucose level,increased blood flow to the heart and muscles,decreased blood flow to the skin and digestive organs.

3.a) Pheromones
   b)To ensure the availability of food.
   Signalling dangers.

4.a) Progesterone
   b) Ovary

5. Hypothalamus secretes releasing hormone. The releasing hormone stimulates the anterior lobe of pituitary gland. The pituitary gland secretes TSH. TSH stimulates thyroid gland to increase thyroxine production.

6. a) Progesterone.
It is essential for regulation of menstrual cycle, preperation of the endometrium for implantation of embryo and maintenance of pregnancy.
b) ovary

7. Prolactin & Oxytocin
Posterior lobe of pituitary.

8. Blurred vision – Retinopathy
Damage to retinal blood vessels due to high glucose.

Numbness in feet – Neuropathy
Nerve damage caused by uncontrolled diabetes.

9. a) Auxin
b) Inhibits lateral bud growth and regulate tropic movements.

10. Hyperthyroidism

In this condition ,the thyroid gland secretes excess thyroxine hormone. This increases the metabolic rate of the body,leading to weight loss.The excess hormone also stimulates the heart,causing an increased heart beat,and over-stimulates the nervous system,resulting in nervousness.

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11. a) Thymus gland
b) The thymus gland is essential for the maturation of T – Lymphocytes..If the gland is congenitally absent,T cells do not mature properly. As a result ,the child will have a weakened immune system, reduced cell – mediated immunity,and will be highly susceptible to infections.

12. a) Melatonin
b) Using a mobile phone late at night delays sleep and reduces sleep quality.
c) The blue light emitted from mobile screen suppresses melatonin secretion.
Melatonin is the hormone responsible for regulating the sleep – wake cycle. When its secretion is reduced ,the body's internal clock is disrupted,making it harder to fall asleep and maintain deep sleep.

13. a) Adrenal gland.
Epinephrine and Norepineprine.

b) They increase the heart rate,raise blood pressure and increase breathing rate. These hormones also increase blood glucose levels by breaking down glycogen,supplying more energy to muscles. Blood flow to vital organs like the brain

and muscles is increased,helping the body to respond quickly and effectively during an emergency.

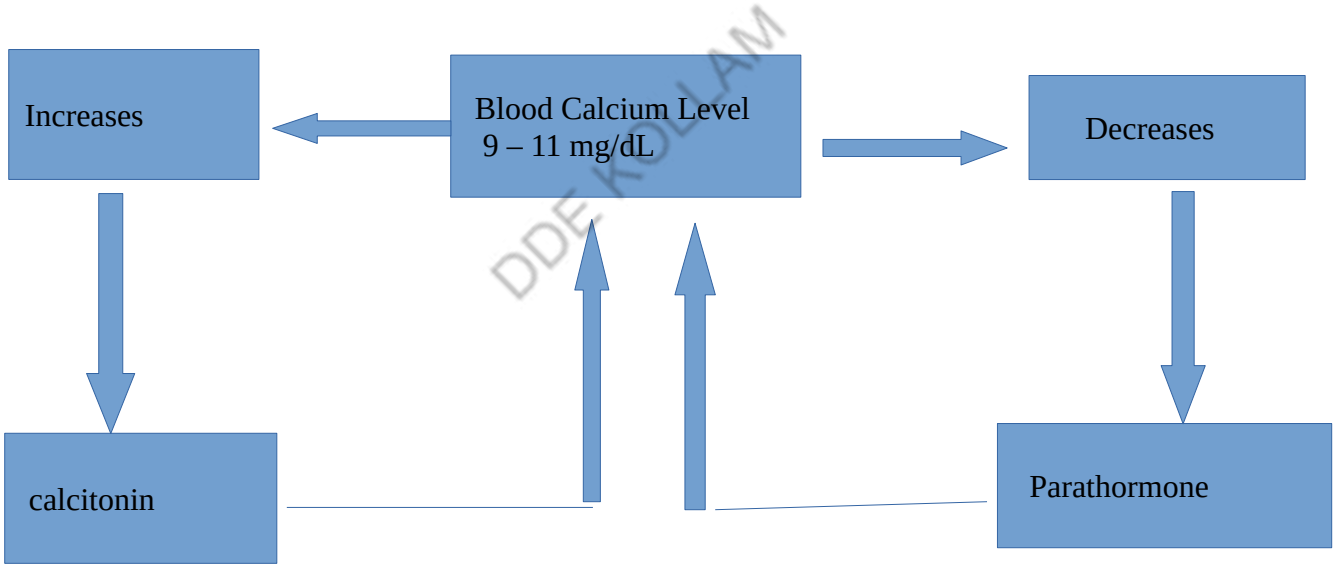
14. a) A – Hypothalamus
    b) Somatotropin , Prolactin
    c)Thyroid gland

15. a) Yes. Flowering is related with length of the day and night (photoperiodism).
    b)Leaves perceive the length of day and night through phytochrome.

16. a) Follicular cells of thyroid gland
    b) Development of brain in newborn babies and children.
    Helps to maintain the health of the heart,skin,and reproductive system

17. a) To increase the size of fruit in grapevines and promote stem elongation in sugarcane.
    b) Ethylene is used to make fruits ripen uniformly for harvest at the same time.

18. a)Insulin
    b)




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19. a) If diabetes is not controlled,it can lead to heart diseases,stroke,renal diseases and vision problems.

b) Type 1 diabetes is the condition in which insulin production capacity is lost due to the destruction of beta cells.

20. a) Acromegaly

b) cause – The production of somatotropin increases after growth phase.

Symptoms – Excessive growth of body parts such as palm,foot and jaws.

21.

| SUMMER | WINTER |
|---|---|
| Production of ADH increases | Production of ADH decreases |
| Kidney increases the reabsorption of water | Kidney decreases the reabsorption of water |
| Production of urine decreases | Production of urine increases |

22. a) Glucagon

b) When the level of glucose in blood increases above normal level,the beta cells(Y) of pancreas secrete insulin (Z).Insulin speeds up the entry of glucose into cells and promotes the conversion of glucose into glycogen in the liver. This action reduces the blood glucose level.

c) Diabetes mellitus

Excessive thirst,hunger,frequent urination,weight loss and fatigue are its symptoms.

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SMART A PLUS

# CHAPTER 5
# IMMUNITY AND HEALTHCARE

**1. Find the correct answer and write it down.**
**a) Lymphocytes are primarily responsible for acquired immunity.**
**b) Red blood cells play a major role in disease resistance.**
**c) Immunity that develops after birth is acquired immunity.**
**1. a and b are correct.** **3. a and b are wrong, c is correct.**
**2. a and c are correct, b is wrong.** **4. All are correct.**

**2. Read the statement and reason given below and choose the appropriate answer.**
**Statement:**
**There are diseases that occur during a person's lifetime and those that are hereditary.**
**Reason:**
**Defects in the immune system, genetic defects, and unhealthy lifestyles disrupt the body's healthy state.**
**a. Statement is correct, reason is wrong.**
**b. Statement is wrong, reason is correct.**
**c. Both statement and reason are correct.**
**d. Both statement and reason are wrong.**

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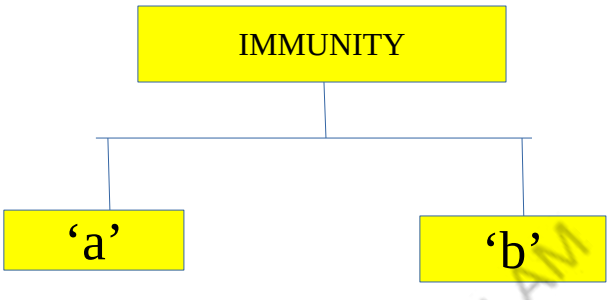
**3. Read the statements below and find the appropriate answer.**
**Statement 1: Microorganisms are the cause of most infectious diseases.**
**Statement 2: Environmental factors, pathogens, and lifestyle etc., cause acquired diseases.**
**a. Statement 1 is correct, 2 is wrong.**
**b. Both statements are wrong.**
**c. Both statements are correct.**
**d. Statement 1 is wrong, 2 is correct.**

**4. Statement 1: Ringworm is a viral disease.**
**Statement 2: Candidiasis is a fungal disease.**
**a. Statement 1 is correct, 2 is wrong.**
**b. Both statements are wrong.**
**c. Both statements are correct.**
**d. Statement 1 is wrong, 2 is correct.**
**5. Correct the wrong statement regarding antibodies if necessary.**
**a. The antibody present in the blood determines a person's blood group.**
**b. Antibodies are found in the plasma.**
**c. Antibodies are found on the surface of red blood cells.**

**6. Statement : In an inflammatory response, the capillary walls dilate.**
**Reason: Monocytes and neutrophils reach the site of the wound.**
**a. Statement is wrong, reason is correct.**
**b. Both are wrong.**
**c. Both are correct.**
**d. Statement is correct, reason is wrong.**

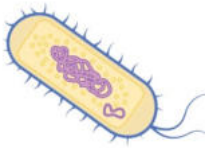
**7. Pathogens and pollen are examples of antigens. List two other examples .**

**8. Write 'a' and 'b' represent in the given chart.**

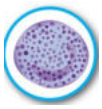



**9. Even after some wounds heal, a scar remains. Evaluate this statement and justify your opinion.**

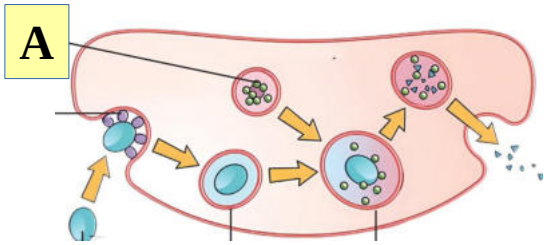
**10. a. Identify the image.**
**b. Name two diseases caused by this organism.**



**11. Arrange columns B and C according to column A.**

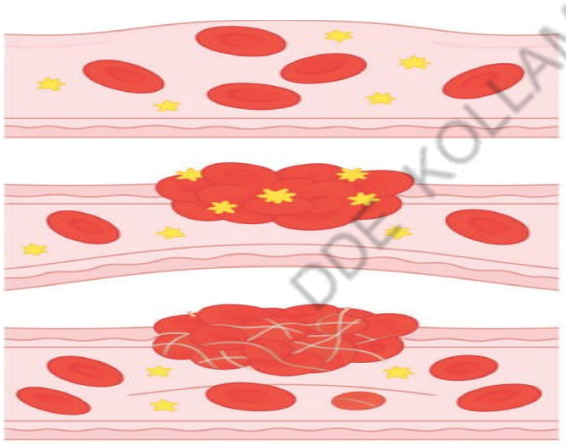
| A | B | C |
|---|---|---|
|  | **Eosinophil** | **Produce histamine** |
|  | **Neutrophil** | **destroys parasites** |
|  | **Basophil** | **It engulfs and destroys bacteria** |

**12. a. Write the function of 'A' in the figure.**
**b. Prepare a flowchart related to phagocytosis.**




**13. Pyrogens are chemical substances produced by white blood cells, play a major role in the immune system. Evaluate this and answer the questions.**

**a. What is the importance of pyrogens?**
**b. What is the benefit of an increase in body temperature?**
**14. An illustration related to the immune system is given below.**
**What are the processes involved in this action?**


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**15. Natural immunity alone may not always be enough to protect the body.**
**a. In such cases, what is the artificial immunity method used to make the body ready ?**
**b. How do these work in the body?**

**16. Categorize the following under appropriate headings.**
**a. Identifies and destroys cancer cells.**
**b. Produces antibodies.**
**c. Formed and matured in the bone marrow.**
**d. Matured in the Thymus gland.**

**17. Complete the table.**

| Vaccine | Components of vaccine | Preventable diseases |
|---|---|---|
| MMR | **B** | Measles, Mumps, Rubella (German measles) |
| **A** | Contains dead pathogens | polio |
| HPV vaccine | Contains parts of pathogens | **D** |
| COVID- 19 vaccine | **C** | Covid 19 |

**18. "Cough lasting for more than two weeks, fever, loss of appetite, and weight loss are symptoms of a disease."**

**a. Identify the disease.**
**b. Explain the main prevention and treatment methods for this.**

**19. a. Which disease is caused by Leptospira?**
**b. Explain how it spreads.**

**20. "Attacks T-lymphocytes and destroys them by multiplying using the genetic mechanism."**
**a. Identify the disease? Explain the mode of transmission and treatment.**

**21. "Cancer is a condition where cells multiply through abnormal and uncontrolled cell division and spread to other parts of the body." (It is a quote from a cancer awareness seminar).**

**a. Explain the causes and treatment methods for this disease.**

**22. Analyze the news report and answer the questions:**

**a. Name the pathogen?**
**b. Mention the symptoms.**




**23. a. Complete the table:**

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| Disease | Pathogen | Vector |
|---|---|---|
| **Malaria** | **-------(A)-------** | **Female Anopheles mosquito** |
| **-------(B)----------** | **Filarial worm** | **--------(C)----------** |

**b. What are the precautions to prevent the diseases mentioned above?**

**24.**

"National Deworming Day: Worm-free children are healthy children."

**a. What is the importance of observing this day?**
**b. Write the symptoms of worm infestation?**

**25.**

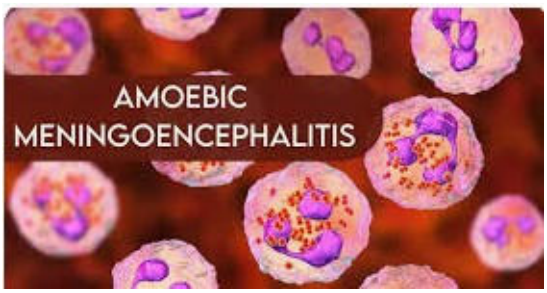



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**a. Which disease does the above stages indicate?**
**b. write the symptoms and treatment of this disease?**

**26. "Use antibiotics only as per a doctor's instruction."**
**a. Do you agree with this statement?**
**b. Justify your answer.**

**27.**




**a. Identify the pathogen responsible for this fatal disease.**
**b. What are the precautions to avoid this disease?**

**28. Which treatment systems are the following statements related to?**

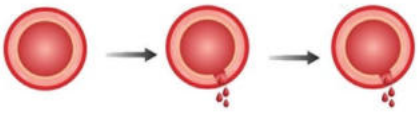
**a. Treatment that originated in India.**
**b. Prevents disease using diluted natural substances.**
**c. Gives importance to the pathogen, diagnosis, and medicine.**

**29. Match the following:**

| |
|---|
| (Nutritional disorders)<br>(Lifestyle diseases)<br><br>(Occupational diseases)<br>(High blood pressure)<br>(Silicosis)<br><br>(Auto immune disorder).<br>(Anaemia)<br><br>(Rheumatoid arthritis) |

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**30. Analyze the diagram and answer the following.**



**a. Identify the disease.**
**b. Write its cause and how it affects the body.**

**31. Write 4 guidelines for blood donors regarding blood donation camps.**



**32. 1.Why is it necessary to consider the blood group during blood transfusion?**
**2. Can a person with A+ve blood group donate blood to a person with B+ve blood group? Give reasons.**

**33. Write the functions of the following substances in plant defence .**
**Enzymes**
**Callose**

**34. Select the Viral diseases from those given below:**

| |
|---|
| **Blight** **Ranikhet disease** **Mosaic disease** **Budrot** **Anthrax** **Quick wilt** **Wilt** **Bunchy top** **Foot and Mouth disease** **Mastitis** |

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**35.List out the things can be done to strengthen our immunity?**

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| Si no | ANSWER KEY | marks |
|---|---|---|
| **1** | **2. a and c are correct, b is wrong.** | **1** |
| **2** | **c. Statement and reason are correct.** | **1** |
| **3** | **c. Both statements are correct.** | 1 |
| **4** | **c. Statement 1 is wrong; Statement 2 is correct.** | 1 |
| **5** | **a. A person's blood group is determined by the antigen present in the blood.**<br><br>**c. Antigens are found on the surface of red blood cells (RBCs).** | 2 |
| **6** | **c.Both statements are correct.** | 1 |
| **7** | **food items, chemicals** | |
| **8** | **a. Innate immunity; b. Acquired immunity.** | 1 |
| **9.** | **If a wound is healed by connective tissue instead of the lost original tissue, a scar remains.** | 1 |
| **10** | **a. Bacteria; b. Tuberculosis; c. Rat fever** | 2 |
| **11** | **a. Neutrophils: Engulf and destroy bacteria.**<br>**b. Basophils: Produce histamine.**<br>**c. Eosinophils: Destroy parasites.** | 3 |
| **12** | **a. Once the phagolysosome is formed, the digestive enzymes within it destroy the pathogen.**<br>**b. 1. Phagocytes attach to the pathogen and form a phagosome. 2. The phagosome joins with a lysosome to form a phagolysosome. 3. Enzymes destroy the pathogen, remnants are expelled** | 1<br><br>3 |
| 13 | **a. Pyrogens trigger the brain to raise the body temperature.**<br>**b. High temperature helps phagocytes destroy pathogens faster and inhibits the growth of pathogens, thus increasing immunity.** | 3 |
| 14 | **a. Tissues and platelets at the wound site produce the enzyme Thromboplastin.**<br>**b. This converts Prothrombin in the plasma to Thrombin.** | 1 |

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<table>
<tr><td></td><td>c. Thrombin converts Fibrinogen into Fibrin.<br>d. Fibrin forms a mesh that traps RBCs and forms a blood clot.</td><td>3</td></tr>
<tr><td>15</td><td>a. Vaccines.<br>b. Vaccines act as antigens, stimulating the immune system to produce antibodies. These antibodies remain in the body to protect against future infections.</td><td>1<br><br><br>2</td></tr>
<tr><td>16</td><td><table>
<tr><td>B-Lymphocytes</td><td>T-Lymphocytes</td></tr>
<tr><td>Produce antibodies</td><td>Recognize and destroy cancer cells</td></tr>
<tr><td>Formed and matured in the bone marrow</td><td>mature in the thymus gland.</td></tr>
</table></td><td>3</td></tr>
<tr><td>17</td><td>a) OPV<br>b) Contains neutralized pathogens; c) Contains genetic parts of the pathogen<br>d) Cervical cancer.</td><td>1<br><br><br>3</td></tr>
<tr><td>18</td><td>a. Tuberculosis; b. BCG Vaccination. Treatment involves antibiotics.</td><td>3</td></tr>
<tr><td>19</td><td>a. Rat fever (Leptospirosis); b. Enters through wounds or mucous membranes (eyes, nose, mouth) from water contaminated with rat urine.</td><td>1<br><br>1</td></tr>
<tr><td>20</td><td>a. AIDS; b. Transmission via sexual contact, shared needles, blood transfusion, or from mother to foetus.</td><td>3</td></tr>
<tr><td>21</td><td>a. Genetic factors, environmental factors, viruses, lifestyle.<br>b. Surgery, Chemotherapy, Radiotherapy, Immunotherapy, Targeted therapy, Hormone therapy.</td><td>1<br><br>2</td></tr>
<tr><td>22</td><td>a. Nipah virus; b. Early symptoms: fever, headache, muscle pain, cough, vomiting. Later: unconsciousness, fits</td><td>1<br>2</td></tr>
<tr><td>23</td><td>A. Plasmodium; B. Mosquito; C. Female Anopheles.<br>b. Avoid mosquito breeding by cleaning stagnant water and practicing "Dry Day."</td><td>1<br>1<br>2</td></tr>
<tr><td>24</td><td>a. To destroy worms and raise awareness about their harmful effects.<br>b. Symptoms: Anaemia, fatigue, lack of concentration, malnutrition, loss of appetite, stomach pain, dizziness, vomiting.</td><td>1<br><br><br>2</td></tr>
</table>

| | | |
|---|---|---|
| 25 | **a. Periodic fever, shivering, headache, vomiting, muscle pain, fatigue.**<br>**b. Anti-malarial drugs are used for treatment.** | 1<br><br>2 |
| 26 | **26: a. Agree.**<br>**b. Overuse or improper use of antibiotics causes side effects (diarrhoea, vomiting, allergies, kidney issues) and leads to antibiotic resistance.** | 1<br><br><br>3 |
| **27** | **a. Naegleria fowleri**<br>**b. This disease can be prevented by avoiding bathing in or washing the face with stagnant and dirty waterbodies.** | 1<br><br>1 |
| **28** | **1. a. Ayurveda, b. Homeopathy, c. Allopathy. 2. Lab tests, assessing symptoms.** | 3<br>1 |
| **29** | **Lifestyle diseases---Diabetes Mellitus**<br>**Occupational diseases----Silicosis**<br>**Autoimmune disorders-----Rheumatoid arthritis**<br>**Nutritional disorders----Anaemia** | 4 |
| **30** | **1. Haemophilia; 2. a. Defect in the gene controlling blood-clotting proteins. b. Excessive bleeding even from small wounds.** | 1<br><br>2 |
| **31** | **1. Ages 18–60; 2. Weight over 50kg; 3.The haemoglobin level in the blood should be above 12.5 g/dL ; 4. People with asthma, fits, heart/kidney disease, or cancer should not donate.** | 4 |
| **32** | **1. If a donor's blood has an antigen not present in the recipient, the recipient's antibodies will attack it, causing blood to clot. Thus, matching blood groups is vital.**<br><br>**2. No, Antigen A in group A blood will react with antibody'a' in B blood group and clotting of blood occurs.** | 4 |
| **33** | **1. Enzymes destroy the cell wall of pathogens.**<br>**2. If pathogens enter through the bark, "Callose" (a polysaccharide) prevents their spread.** | 2 |
| **34** | **Mosaic disease, Quick wilt (of pepper), Bunchy top (of banana), Bud rot (of coconut).** | 2 |
| **35** | **Balanced diet, proper sleep, stress management, regular exercise, good hygiene.** | 3 |

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# UNIT 6
# BIOLOGY AND TECHNOLOGY

**1. The growth of biotechnology accelerated by the discovery of**

**(Cell, Chromosome, DNA, Gene) (score=1)**

**2. From the following items, select the two technologies used for genetic engineering:**

- **Recombinant DNA Technology**
- **DNA Fingerprinting.**
- **Artificial Intelligence**
- **CRISPR-Cas9 technology (score=1)**

**3. A restriction enzyme cuts DNA only at a specific sequence. Why is this specificity important in Recombinant DNA technology?**

**a) Prevents protein formation**
**b) Ensures same cut pattern**
**c) Allows controlled DNA modification**
**d) Reduces DNA replication (score=1)**

**4. The enzyme acting as molecular scissors in CRISPR is**

**a) Cas1**
**b) Cas9**
**c) Ligase**
**d) Helicase (score=1)**

**5. Which of these GMO plants shows pest resistance?**

**a) Soya bean**
**b) Golden rice**
**c) Bt Cotton**
**d) Papaya (score=1)**

**6. CRISPR technology is considered better than recombinant DNA technology because:**

**a) it works without enzymes**
**b) it edits DNA at precise locations**
**c) it requires plasmids**
**d) it used only in humans (score=1)**

**7. A scientist wants to correct a single wrong base pair in a gene. Which technique is the best?**

**a) Traditional recombinant DNA technology**
**b) CRISPR–Cas9**
**c) Fermentation**
**d) PCR (score=1)**

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**8. If guide RNA is incorrectly designed, CRISPR will:**

**a) Cut wrong DNA region**
**b) Fail DNA replication**
**c) Produce insulin**
**d) Destroy plasmid** **(score=1)**

**9. The percentage of DNA that is identical in all humans is approximately**
**a) 95%**
**b) 97%**
**c) 99.9%**
**d) 90%** **(score=1)**

**10. CRISPR requires:**
**a) Ligase + Cas9**
**b) Guide RNA + Cas9**
**c) Ligase + plasmid**
**d) Restriction enzyme + helicase** **(score=1)**

**11. The technology used for the completion of Human Genome Project that launched in 1990:**

**a)Gene Mapping**
**b)Gene Therapy**
**c)Gene editing**
**d) Gene analysis** **(score=1)**

**12. The aim of Human Microbiome Project was,**

**a) Modify human genes**
**b) Identify human microorganisms**
**c) Clone humans**
**d) Study human organs** **(score=1)**

**13. The main use of Bioinformatics is to**

**a) Treat diseases**
**b) Store and analyse biological data**
**c) Clone organisms**
**d) Print organs** **(score=1)**

**14. Identify and name the project indicated by this logo. Mention its importance.** **(score=2)**




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**15. Assertion(A): Human microbiome plays an important role in digestion.**
**Reason (R): Microorganisms help in breaking down food and producing vitamins.**

**Read the above statement(A) and reason(R) , and select the most suitable one from the following.**

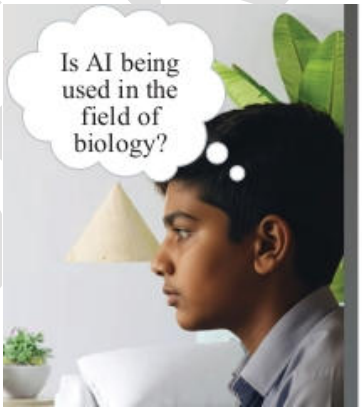
**A. Both A and R are true and R explains A**
**B. Both true but R does not explain A**
**C. A true, R false**
**D. A false, R true** **(Score 1)**

**16. Analyse the given statement and answer the questions.**

**"DNA fingerprinting is a technology used to identify individuals by analysing the sequence of nucleotides."**

**a) How is it possible to identify individuals using DNA fingerprinting technology?**
**b) Write any two other uses of this technology.** **(Score 3)**

**17. Write any three potential applications of artificial intelligence (AI) in the fields of biology?**

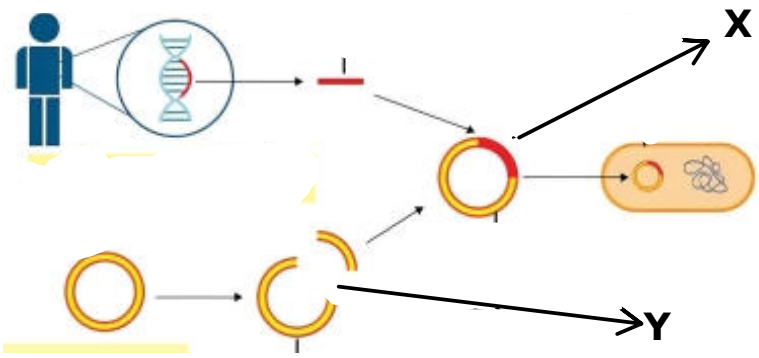



**(score 3)**

**18. Numerous biological techniques have been developed to help conserve the ecosystems and biodiversity. Name and explain any three modern techniques used to protect the environment.** **(Score 3)**

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**19.Analyse the diagram and answer the questions.**




**a) Identify and name the process? How does this process bring about changes in the genetic structure of organisms?**

**b)Write the names of the enzymes used in the processes X and Y.** **(Score 4)**

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# 6-BIOLOGY AND TECHNOLOGY

**ANSWER KEY**

**1. DNA**

**2. Recombinant DNA Technology and CRISPR-Cas9 Technology**

**3. Allows controlled DNA modification**

**4. Cas9**

**5. Bt Cotton**

**6. It edits DNA at precise locations**

**7. CRISPR–Cas9**

**8. Cut wrong DNA region**

**9. 99.9%**

**10. Guide RNA + Cas9**

**11. Gene Mapping**

**12. Identify human microorganisms**

**13. Store and analyse biological data**

**14. Project: Human Microbiome Project**
**Importance: It identifies and studies the microorganisms living in the human body and their role in health and disease.**

**15. A. Both A and R are true and R explains A**

**16. DNA Fingerprinting**
**a) Individuals can be identified because every person (except identical twins) has a unique DNA sequence. By comparing DNA banding patterns obtained after electrophoresis, individuals can be distinguished.**
**b) Other uses:**
- **To identify criminals in forensic investigations.**
- **To determine paternity or family relationships.**
- **To identify remains of individuals in disasters.**

**17. Applications of Artificial Intelligence (AI) in Biology**
- **Disease diagnosis and prediction using medical imaging and data analysis.**
- **Drug discovery and development.**
- **Genome analysis and protein structure prediction.**
- **Monitoring and management of ecosystems. (any three)**

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**18. Modern Techniques for Environmental Protection**

- **Tissue culture: Used to produce large numbers of disease-resistant and high-yielding plants.**
- **Cryopreservation: Preserving endangered species' gametes or embryos at very low temperatures.**
- **DNA barcoding: Identifying and cataloguing species for biodiversity conservation.**
- **GMOs for pollution control: Genetically modified microbes used for bioremediation. (any three with explanation)**

**19. Recombinant DNA Technology**

**a) Process: Recombinant DNA technology (genetic engineering).**

**•The process involves combining DNA from two different sources (e.g., human gene inserted into bacterial plasmid).**

**•It introduces new genes into an organism, changing its genetic structure.**

**b) Enzymes used:**

**•X: Restriction enzyme (cuts DNA at specific sites).**

**•Y: Ligase (joins DNA fragments).**

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